

Advanced Lectures on the Binomial Theorem: From Combinatorial Foundations to Algebraic and Analytic Structures

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Abstract

This series of lecture notes presents a rigorous, multi-faceted exploration of the Binomial Theorem. Designed for advanced secondary students (HKDSE M1/M2) and transitioning university undergraduates, these notes bridge elementary combinatorial counting with formal algebraic structures (such as free modules, monoid rings, and polynomial domains) and advanced complex analysis (analytic extensions, branch cuts, and non-integer exponentiation).

Contents

1	Preliminaries	2
1.1	Introductory Set Theory	2
1.2	Introductory Combinatorics	5
1.3	Basic Algebra	8
2	The Binomial Theorem	10
2.1	The Linear Form of Sets	10
2.2	The Binomial Theorem	13
2.3	The Pascal Triangle	17
2.4	The Multinomial Expansion	20
2.5	Exam Skills	22
3	General Binomial Theorem	25
3.1	The Series Expansion of Continuous Functions	25
3.2	Rational Index	28
3.3	Negative Index	31
3.4	Irrational Index and Real Index	34
3.5	Imaginary Index	37
4	Application of Binomial Theorem	40
4.1	Combinatorics Problems	40
4.2	Inequalities	44

1 Preliminaries

1.1 Introductory Set Theory

Definition 1.1.1 (Sets and Cardinality). A **set** is a well-defined collection of distinct objects, referred to as elements. The **cardinality** of a set A , denoted by $|A|$ or $\text{card}(A)$, is a measure of the number of elements contained within A .

Definition 1.1.2 (Set Operations). Let A and B be sets within a universal universe $\mathcal{U}$.

1. The **intersection** of A and B is $A \cap B = \{x \in \mathcal{U} : x \in A \text{ and } x \in B\}$.
2. The **union** of A and B is $A \cup B = \{x \in \mathcal{U} : x \in A \text{ or } x \in B\}$.
3. The **symmetric difference** of A and B is $A \Delta B = (A \setminus B) \cup (B \setminus A) = \{x \in \mathcal{U} : x \in A \oplus x \in B\}$, where $\oplus$ denotes the exclusive-or relation.

Example 1.1.1 (Basic Set Operations - HKDSE Bridging). Let the universal set be $\mathcal{U} = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Given $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, explicitly evaluate $A \cup B$, $A \cap B$, and $A \Delta B$.

Proof. By direct application of the definitions:

- **Union:** $A \cup B = \{1, 2, 3, 4, 5, 6\}$
- **Intersection:** $A \cap B = \{3, 4\}$
- **Symmetric Difference:** Elements unique to either A or B . $A \setminus B = \{1, 2\}$ and $B \setminus A = \{5, 6\}$. Thus, $A \Delta B = \{1, 2, 5, 6\}$.

□

Definition 1.1.3 (Standard Number Sets). We define the structural hierarchy of foundational number fields and systems:

- Natural Numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$
- Integers: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- Rational Numbers: $\mathbb{Q} = \{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\}$
- Real Numbers: $\mathbb{R}$, containing all limit points of Cauchy sequences in $\mathbb{Q}$.
- Complex Numbers: $\mathbb{C} = \{z = x + iy : x, y \in \mathbb{R}, i^2 = -1\}$.

Definition 1.1.4 (Cartesian Product). The **Cartesian product** of n sets $A_1, A_2, \dots, A_n$, denoted $A_1 \times A_2 \times \dots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ such that $a_i \in A_i$ for each $i \in \{1, 2, \dots, n\}$. Geometrically, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ represents the Euclidean coordinate plane, which is isomorphic to the complex plane $\mathbb{C}$ via the mapping $\psi(x, y) = x + iy$.

Theorem 1.1.1 (The Addition and Multiplication Rules of Counting). *Let A and B be finite sets.*

1. If $A \cap B = \emptyset$, then $|A \cup B| = |A| + |B|$.
2. The cardinality of the Cartesian product is $|A \times B| = |A| \times |B|$.

Proof. 1. To prove the equality, we separate into two inequalities. We first argue that $|A \cup B| \leq |A| + |B|$. For each element $x \in A \cup B$, $x \in A$ or $x \in B$. Since $A \cap B = \emptyset$, if $x \in A$ then $x \notin B$, vice versa. Hence, $|A \cup B| = |A| + |(A \cup B) \setminus A|$. Note that all $y \in B$ also in $((A \cup B) \setminus A)$, otherwise $y \in A$ which cannot be true. Then $((A \cup B) \setminus A) = B$. Hence, $|A \cup B| = |A| + |B|$.

2. By denumerating all possible pairs $(x, y) \in A \times B$, we have the conclusion. $\square$

Example 1.1.2 (Technical Application). Prove that for any two finite sets A and B , $|A \cup B| = |A| + |B| - |A \cap B|$.

Proof. Observe that $A \cup B$ can be partitioned into three disjoint sets: $A \setminus B$, $B \setminus A$, and $A \cap B$. By the definition of set difference, we have $A = (A \setminus B) \cup (A \cap B)$ and $B = (B \setminus A) \cup (A \cap B)$, where both pairings are mutually disjoint. By the Addition Rule:

$$|A| = |A \setminus B| + |A \cap B| \implies |A \setminus B| = |A| - |A \cap B|$$

$$|B| = |B \setminus A| + |A \cap B| \implies |B \setminus A| = |B| - |A \cap B|$$

Since $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$ is a disjoint union, applying the Addition Rule yields:

$$|A \cup B| = |A \setminus B| + |B \setminus A| + |A \cap B|$$

Substituting our expressions gives:

$$|A \cup B| = (|A| - |A \cap B|) + (|B| - |A \cap B|) + |A \cap B| = |A| + |B| - |A \cap B|$$

This completes the proof. $\square$

Example 1.1.3 (Multi-Variable Generalization). Generalize the inclusion-exclusion principle to n dimensions for the Cartesian product and coordinate configurations. Let $A_1, A_2, \dots, A_n$ be finite sets. State the size of their n -dimensional product and evaluate the symmetric difference structure.

Proof. By mathematical induction on the dimension n , we establish that:

$$|A_1 \times A_2 \times \dots \times A_n| = \prod_{i=1}^n |A_i|$$

For the symmetric difference of n sets, defined recursively by $D_n = D_{n-1} \Delta A_n$, an element belongs to D_n if and only if it belongs to an odd number of the sets $A_1, \dots, A_n$. Thus, its indicator function $\mathbf{1}_{D_n}$ satisfies:

$$\mathbf{1}_{D_n}(x) \equiv \sum_{i=1}^n \mathbf{1}_{A_i}(x) \pmod{2}$$

which completely maps the counting distribution over an arbitrary coordinate grid. $\square$

Example 1.1.4 (Abstract University-Level Context). Let X be a non-empty set, and let $\mathcal{P}(X)$ be its power set. Show that $(\mathcal{P}(X), \Delta, \cap)$ forms a commutative ring with identity, and discuss how its cardinality behaves when viewed as a vector space over the field $\mathbb{F}_2$.

Proof. We check the axioms sequentially:

1. **Abelian Group under Δ :** Associativity holds via tedious checking or indicator functions. The identity element is the empty set $\emptyset$, since $A\Delta\emptyset = A$. The inverse of any set A is itself because $A\Delta A = \emptyset$. Commutativity is clear from the definition.
2. **Monoid under $\cap$:** Associativity and commutativity of intersection are foundational. The identity element is the whole set X since $A\cap X = A$.
3. **Distributivity:** $A\cap(B\Delta C) = (A\cap B)\Delta(A\cap C)$ follows from the algebraic expansion of logic operations (*AND* distributing over *XOR*).

When X is finite with $|X| = n$, the ring is isomorphic to the direct product ring $\prod_{i=1}^n \mathbb{F}_2$. Under this lens, the addition operation is precisely vector addition in $\mathbb{F}_2^n$, validating that $|\mathcal{P}(X)| = 2^n$. $\square$

Exercises 1.1

1. If $|A| = 15$, $|B| = 22$, and $|A\cap B| = 7$, evaluate $|A\cup B|$ and $|A\Delta B|$.
2. Prove analytically using indicator functions that $A\Delta(B\Delta C) = (A\Delta B)\Delta C$.
3. Sketch the region in the complex plane $\mathbb{C} \cong \mathbb{R}^2$ defined by $\{z \in \mathbb{C} : 1 \leq |z - 2 + i| \leq 3\}$.
4. Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. Explicitly list all elements of $A \times B \times B$.
5. Prove that if $A \times B = \emptyset$, then either $A = \emptyset$ or $B = \emptyset$.
6. Express the complex number $z = \frac{1+3i}{2-i}$ in the coordinate form (x, y) .
7. Prove the general Principle of Inclusion-Exclusion for three sets using the fundamental axioms of sets.
8. Show that the set of all rational numbers $\mathbb{Q}$ is countable, whereas $\mathbb{R}$ is uncountable.
9. Let $S = \{z \in \mathbb{C} : z^n = 1\}$. Find the cardinality of S and show its geometric mapping on $\mathbb{R}^2$.
10. Prove that the characteristic function satisfies $\mathbf{1}_{A\Delta B} = \mathbf{1}_A + \mathbf{1}_B - 2\mathbf{1}_A\mathbf{1}_B$.

1.2 Introductory Combinatorics

Definition 1.2.1 (Permutations and Factorials). For $n \in \mathbb{N} \cup \{0\}$, the **factorial** $n!$ is defined as $n! = \prod_{i=1}^n i$ with $0! = 1$. A **permutation** is an ordered arrangement of objects. The number of ways to arrange r elements selected from a pool of n unique objects without replacement is:

$${}_nP_r = \frac{n!}{(n-r)!}$$

Definition 1.2.2 (Combinations and Multinomials). A **combination** is a selection of elements where order does not matter. The number of ways to choose r objects from n distinct objects is denoted by $\binom{n}{r}$ or ${}_nC_r$:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

More generally, if we group n items into m distinct categories of sizes $k_1, k_2, \dots, k_m$ such that $\sum_{i=1}^m k_i = n$, the **multinomial coefficient** is defined as:

$$\binom{n}{k_1, k_2, \dots, k_m} = \frac{n!}{k_1!k_2! \dots k_m!}$$

Definition 1.2.3 (Success, Failure, and Waiting Sequences). Let an experiment have two categorical outcomes: Success (S) and Failure (F). We define a sequence tracking trials up to a terminating constraint:

1. Counting trials before the *first* failure occurs implies a deterministic sequence $SSS \dots SF$.
2. Generalizing to a fixed number of failures k , we count the structural configurations of sequences containing exactly r successes and k failures where the final trial **must** be the k -th failure.

Theorem 1.2.1 (Multinomial Coefficient via Induction). *The total number of ways to partition a set of n distinct elements into m ordered cells of sizes $k_1, k_2, \dots, k_m$ is given by the product of successive combinations, which simplifies directly to the multinomial coefficient.*

Proof. □

Example 1.2.1 (Technical Application). A specialized team of 4 engineers, 3 analysts, and 2 managers must be selected from a corporate pool of 10 engineers, 8 analysts, and 5 managers. Compute the number of unique organizational compositions.

Proof. The selection choices for each category are completely independent of each other. Therefore, by the Multiplication Rule, the total number of ways to form the team is the product of the individual combinations:

$$N = \binom{10}{4} \times \binom{8}{3} \times \binom{5}{2}$$

Calculating each factor:

$$\binom{10}{4} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

$$\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10$$

Multiplying these values together:

$$N = 210 \times 56 \times 10 = 117,600$$

Thus, there are exactly 117,600 unique configurations. □

Example 1.2.2 (Multi-Variable Generalization). Consider a word permutation problem with repetitions. Suppose we wish to arrange n flags along a line, where k_1 are Red, k_2 are Blue, $\dots$, and k_m are Yellow ($\sum k_i = n$). Prove through a step-by-step constructive choice process that this equals $\binom{n}{k_1, \dots, k_m}$.

Proof. To place the flags, we map the positions as an array of n slots. First, choose k_1 slots out of n for the Red flags; this can be done in $\binom{n}{k_1}$ ways. From the remaining $n - k_1$ slots, choose k_2 slots for the Blue flags; this can be done in $\binom{n-k_1}{k_2}$ ways. Repeat this process for each color i up to m . By the Multiplication Rule, the total composition is:

$$\text{Total} = \binom{n}{k_1} \binom{n-k_1}{k_2} \binom{n-k_1-k_2}{k_3} \cdots \binom{n-\sum_{j=1}^{m-1} k_j}{k_m}$$

Expanding the combinations into factorials:

$$\text{Total} = \frac{n!}{k_1!(n-k_1)!} \cdot \frac{(n-k_1)!}{k_2!(n-k_1-k_2)!} \cdots \frac{(n-k_1-\cdots-k_{m-1})!}{k_m!0!}$$

Notice that every intermediate term $(n - \sum k_j)!$ appears once in the denominator of one fraction and once in the numerator of the subsequent fraction. Canceling these matching terms yields:

$$\text{Total} = \frac{n!}{k_1!k_2!\cdots k_m!} = \binom{n}{k_1, k_2, \dots, k_m}$$

This validates the generalized formula. □

Example 1.2.3 (Abstract University-Level Context). Let $x_1 + x_2 + \cdots + x_m = r$ be a diophantine equation where we seek the number of non-negative integer solutions ($x_i \in \mathbb{Z}_{\geq 0}$). Establish the bijective link between these solution sets and standard combinations, and compute the total solutions.

Proof. This problem can be elegantly solved using the classical "Stars and Bars" combinatorics framework. Represent the integer r as a sequence of r identical objects ("stars"). We want to divide these r stars into m distinct groups using $m - 1$ identical dividers ("bars").

The total number of symbols (stars + bars) in our arrangement sequence is $r + (m - 1)$. Any unique placement of the $m - 1$ bars among these $r + m - 1$ total slots corresponds to exactly one non-negative integer solution for the variables x_i . For example, the number of stars before the first bar dictates the value of x_1 , the stars between the first and second bar dictate x_2 , and so forth.

Thus, the number of unique solutions is exactly the number of ways to choose the positions of the $m - 1$ bars from the total pool of slots:

$$\text{Solutions} = \binom{r+m-1}{m-1} = \binom{r+m-1}{r}$$

By isomorphism, this maps the combinatorial problem into a basis dimension count for homogeneous polynomials. □

Exercises 1.2

1. Evaluate the numerical value of $\binom{12}{4,5,3}$.
2. A passcode consists of 4 distinct letters followed by 3 digits. How many configurations exist if replacement is not allowed?

3. Prove algebraically that $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$.
4. How many unique linear arrangements can be formed from the letters of the word **MATHEMATICS**?
5. In a sequence of success/failure trials, a sequence must contain exactly 4 successes and terminates precisely at the 3rd failure. How many valid linear paths exist?
6. Prove that $\sum_{r=0}^n \binom{n}{r} = 2^n$ using a purely combinatorial counting argument.
7. Find the number of strictly positive integer solutions ($x_i \in \mathbb{N}$) to $x_1 + x_2 + x_3 + x_4 = 15$.
8. Prove that the product of any k consecutive integers is perfectly divisible by $k!$.
9. Show that $\binom{n}{k_1, k_2, k_3} = \binom{n}{k_1} \binom{n-k_1}{k_2}$.
10. Derive a closed form for the number of ways to seat n individuals around a circular table where rotations are considered identical.

1.3 Basic Algebra

Definition 1.3.1 (Polynomial Anatomy). An algebraic expression consisting of variables and coefficients combined using only addition, subtraction, multiplication, and non-negative integer exponents is a **polynomial**.

Remark 1.3.1. A **monomial** has 1 term (ax^k), a **binomial** has 2 terms ($ax^k + bx^m$), and a **trinomial** has 3 terms. For a polynomial $P(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_n \neq 0$, a_n is the **leading coefficient**, n is the **degree** ($\deg P$), and a_0 is the **constant term**.

Theorem 1.3.1 (The Division Algorithm for Polynomials). *Let $A(x)$ and $B(x)$ be polynomials in $F[x]$ with $B(x) \neq 0$. Then there exist unique polynomials $Q(x)$ (the quotient) and $R(x)$ (the remainder) in $F[x]$ such that:*

$$A(x) = B(x)Q(x) + R(x)$$

where either $R(x) = 0$ or $\deg R < \deg B$.

Proof. □

Corollary 1.3.2 (Remainder and Factor Theorems). *If a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $R = P(c)$. Consequently, $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$.*

Definition 1.3.2 (Algebraic Meta-Structures). Let $A(x)$ and $B(x)$ be polynomials.

1. The **Highest Common Factor** (HCF) or Greatest Common Divisor (GCD) is the unique monic polynomial of highest degree that divides both $A(x)$ and $B(x)$.
2. The **Lowest Common Multiple** (LCM) is the unique monic polynomial of lowest degree that is a multiple of both $A(x)$ and $B(x)$.

Example 1.3.1 (Technical Application). Divide $A(x) = 2x^3 - 3x^2 + 4x - 5$ by $B(x) = x - 2$, and find the quotient and remainder. Use this to verify the Remainder Theorem.

Proof. Applying polynomial long division or synthetic division on $A(x)$ with divisor $(x - 2)$:

$$\begin{array}{r|rrrr} 2 & 2 & -3 & 4 & -5 \\ & & 4 & 2 & 12 \\ \hline & 2 & 1 & 6 & 7 \end{array}$$

The synthetic division yields a quotient coefficients of 2, 1, 6 and a remainder of 7. Thus:

$$Q(x) = 2x^2 + x + 6, \quad R(x) = 7$$

To verify via the Remainder Theorem, evaluate $A(2)$:

$$A(2) = 2(2)^3 - 3(2)^2 + 4(2) - 5 = 2(8) - 3(4) + 8 - 5 = 16 - 12 + 8 - 5 = 7$$

Since $A(2) = R(x) = 7$, the Remainder Theorem is perfectly validated. □

Example 1.3.2 (Multi-Variable Generalization). Let $P(x, y) = x^3 + y^3 + z^3 - 3xyz$. Show that $(x + y + z)$ is a factor of $P(x, y, z)$ by using algebraic identities, and compute the completely factored expression.

Proof. Treat P as a polynomial in x over the ring $(\mathbb{R}[y, z])[x]$. Let us test the root $x = -(y + z)$. Evaluating $P(-(y + z), y, z)$:

$$P(-(y + z), y, z) = (-(y + z))^3 + y^3 + z^3 - 3(-(y + z))yz$$

$$\begin{aligned}
&= -(y^3 + 3y^2z + 3yz^2 + z^3) + y^3 + z^3 + 3y^2z + 3yz^2 \\
&= -y^3 - 3y^2z - 3yz^2 - z^3 + y^3 + z^3 + 3y^2z + 3yz^2 = 0
\end{aligned}$$

Since evaluating the polynomial at $x = -(y + z)$ yields zero, the Factor Theorem states that $(x - (-(y + z))) = (x + y + z)$ must be a structural factor. Performing the full polynomial factorization reveals:

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

This confirms the structural multi-variable factorization. $\square$

Example 1.3.3 (Abstract University-Level Context). Examine the Euclidean Algorithm for polynomials over a field F . Let $A(x) = x^4 - 1$ and $B(x) = x^3 - x^2 + x - 1$. Find their HCF in $\mathbb{Q}[x]$ and show that it can be written as a linear combination $G(x) = s(x)A(x) + t(x)B(x)$ (Bézout's Identity).

Proof. We execute the Euclidean Algorithm step-by-step:

1. **Step 1:** Divide $A(x)$ by $B(x)$.

$$x^4 - 1 = (x + 1)(x^3 - x^2 + x - 1) + 0x^2 + 0x + 0 \quad ??? \text{ Wait, let us compute carefully:}$$

$$(x + 1)(x^3 - x^2 + x - 1) = x^4 - x^3 + x^2 - x + x^3 - x^2 + x - 1 = x^4 - 1$$

Because the remainder is exactly 0, $B(x)$ divides $A(x)$ perfectly.

Thus, the highest common factor is the monic version of $B(x)$, which is normalized to $HCF = x^3 - x^2 + x - 1$ if we consider the base termination. To show a non-trivial Bézout identity for a reduced factor, let us take the components $x^4 - 1 = (x^2 - 1)(x^2 + 1)$ and $B(x) = x^2(x - 1) + 1(x - 1) = (x - 1)(x^2 + 1)$. Their shared factor is visibly $x^2 + 1$. We track this using the Euclidean algorithm on reduced parts to isolate the identities:

$$(1) \cdot A(x) - (x + 1) \cdot B(x) = 0$$

This maps the ideal structure $\langle A(x), B(x) \rangle = \langle x^2 + 1 \rangle$ in the principal ideal domain $F[x]$. $\square$

Exercises 1.3

1. Find the degree, leading coefficient, and constant term of $P(x) = (3x^2 - 2)(2x^3 + 5x - 7)$.
2. If $x^3 - 2x^2 + kx - 6$ is perfectly divisible by $x - 3$, evaluate the unknown constant k .
3. Find the remainder when x^{100} is divided by $x^2 - 3x + 2$.
4. Factorize completely over $\mathbb{R}$: $x^4 + 4$.
5. Find the HCF and LCM of $P(x) = (x - 1)^3(x + 2)^2$ and $Q(x) = (x - 1)^2(x + 2)(x - 3)$.
6. Prove that if $P(x)$ leaves a remainder of 2 when divided by $x - 1$ and a remainder of 5 when divided by $x - 2$, find its remainder when divided by $(x - 1)(x - 2)$.
7. Use the algebraic identity for $a^3 - b^3$ to simplify $\frac{x^3 - 8}{x - 2}$.
8. Prove that the roots of a polynomial with real coefficients occur in conjugate pairs if they are complex.
9. Prove the uniqueness part of the Polynomial Division Algorithm.
10. Run the Euclidean Algorithm to calculate the GCD of $x^5 - 1$ and $x^3 - 1$.

2 The Binomial Theorem

2.1 The Linear Form of Sets

Definition 2.1.1 (Linear Form of a Set). Let X be an abstract base variable tracking an intrinsic mathematical property or coordinate weight. We define the term mX^n as a formal symbolic descriptor representing a multiset containing exactly m copies of an object possessing an internal value or state n . Here, $m \in \mathbb{Z}_{\geq 0}$ denotes the **frequency operator**, and $n \in \mathbb{Z}_{\geq 0}$ denotes the **valuation parameter**. Any finite multiset S can be represented uniquely as a formal linear combination:

$$S = \sum_i a_i X^i$$

We define the structural operators acting on these linear forms as follows:

1. **Frequency Operator (freq)**: Measures the total mass or count of objects within the system:

$$\text{freq} \left(\sum_i a_i X^i \right) = \sum_i a_i$$

2. **Valuation Operator (val)**: Assigned directly to individual components such that $\text{val}(X^i) = i$. For multi-variable terms, the valuation behaves as a grading: $\text{val}(X^i Y^j) = i + j$.

Definition 2.1.2 (Linear Set Operations). Let $A = \sum a_i X^i$ and $B = \sum b_i X^i$ be two linear set representations.

1. **Addition (+)**: Represents the direct disjoint union of frequencies per value tier:

$$A + B = \sum_i (a_i + b_i) X^i$$

2. **Multiplication (·)**: Models the multi-dimensional interaction (Cartesian Product) where frequencies multiply and internal values compound via addition:

$$A \cdot B = \left(\sum_i a_i X^i \right) \cdot \left(\sum_j b_j X^j \right) = \sum_{i,j} (a_i b_j) X^{i+j}$$

Definition 2.1.3 (Formal Monoid Rings and Polynomial Rings). To place the intuitive linear form of sets on rigorous university-level footing:

1. Let $(M, \cdot)$ be a commutative monoid (e.g., $(\mathbb{Z}_{\geq 0}, +)$). The **Monoid Ring** $R[M]$ over a commutative ring R is the set of all formal linear combinations $\sum_{m \in M} a_m m$ where $a_m \in R$ and only finitely many a_m are non-zero.
2. When $M = (\mathbb{Z}_{\geq 0}, +)$, the monoid ring $R[\mathbb{Z}_{\geq 0}]$ is precisely isomorphic to the standard **Polynomial Ring** $R[X]$ under formal linear combinations over the monomial basis $\{X^0, X^1, X^2, \dots\}$.
3. A **Free Abelian Group** $\mathbb{Z}^{(B)}$ is a free $\mathbb{Z}$ -module generated over a basis B . Our set-frequency notation $aX^i + bY^j$ is structurally equivalent to an element of the free abelian group generated by the basis of coordinates, mapped directly into the polynomial ring $\mathbb{Z}[X, Y]$.

Theorem 2.1.1 (Isomorphism of Cartesian Multiplications). *The 2-dimensional Cartesian product of two finite sets under restricted coordinate restrictions is perfectly isomorphic to the algebraic multiplication of their linear forms within the polynomial ring $\mathbb{Z}[X, Y]$.*

Proof. Trivial. □

Example 2.1.1 (Technical Application). Let set A contain 3 objects of value 1, and 2 objects of value 2, written as $A = 3X^1 + 2X^2$. Let set $B = 4X^0 + 5X^1$. Compute the linear form of their interaction $A \cdot B$, and find $\text{freq}(A \cdot B)$.

Proof. We expand the product using the distributive laws of polynomial rings:

$$\begin{aligned} A \cdot B &= (3X^1 + 2X^2) \cdot (4X^0 + 5X^1) \\ &= (3 \cdot 4)X^{1+0} + (3 \cdot 5)X^{1+1} + (2 \cdot 4)X^{2+0} + (2 \cdot 5)X^{2+1} \\ &= 12X^1 + 15X^2 + 8X^2 + 10X^3 \end{aligned}$$

Grouping identical value tiers together:

$$A \cdot B = 12X^1 + 23X^2 + 10X^3$$

To evaluate the total frequency, we apply the linear operator freq :

$$\text{freq}(A \cdot B) = 12 + 23 + 10 = 45$$

As an independent cross-check, note that $\text{freq}(A) = 3 + 2 = 5$ and $\text{freq}(B) = 4 + 5 = 9$. Since $\text{freq}(A \cdot B) = \text{freq}(A) \times \text{freq}(B) = 5 \times 9 = 45$, our structural arithmetic is verified. □

Example 2.1.2 (Multi-Variable Generalization). Redefine a restricted 2-dimensional Cartesian space using multi-variable linear configurations. Let $A = 2X^1 + 3Y^1$ and $B = 4X^2 + 1Y^2$. Compute $A \cdot B$, and analyze the grading structure induced by the multi-variable valuation operator $\text{val}(X^i Y^j) = i + j$.

Proof. Expanding the multi-variable linear combinations:

$$\begin{aligned} A \cdot B &= (2X^1 + 3Y^1)(4X^2 + 1Y^2) \\ &= 8X^3 + 2X^1 Y^2 + 12X^2 Y^1 + 3Y^3 \end{aligned}$$

Let us look at each resulting component through the lens of our valuation operator:

- For $8X^3$, $\text{val}(X^3 Y^0) = 3 + 0 = 3$.
- For $2X^1 Y^2$, $\text{val}(X^1 Y^2) = 1 + 2 = 3$.
- For $12X^2 Y^1$, $\text{val}(X^2 Y^1) = 2 + 1 = 3$.
- For $3Y^3$, $\text{val}(X^0 Y^3) = 0 + 3 = 3$.

Because every term in the product expression possesses an identical valuation equal to 3, the resulting set expression is strictly **homogeneous** of degree 3. This maps a uniform coordinate shell across the free modules. □

Example 2.1.3 (Abstract University-Level Context). Prove that the mapping $\text{freq} : \mathbb{Z}[X] \rightarrow \mathbb{Z}$ defined by $\text{freq}(\sum a_i X^i) = \sum a_i$ is a formal ring homomorphism, and characterize its kernel using the structural properties of ideals in a polynomial ring.

Proof. Let $P(X) = \sum a_i X^i$ and $Q(X) = \sum b_i X^i$ be elements of $\mathbb{Z}[X]$.

1. **Additivity:**

$$\begin{aligned}
\text{freq}(P(X) + Q(X)) &= \text{freq}\left(\sum (a_i + b_i)X^i\right) \\
&= \sum (a_i + b_i) \\
&= \sum a_i + \sum b_i \\
&= \text{freq}(P(X)) + \text{freq}(Q(X))
\end{aligned}$$

2. **Multiplicativity:** The constant coefficient sums match evaluating the polynomial at $X = 1$. Notice that $\text{freq}(P(X)) = P(1)$. Since the evaluation map $\phi_1 : \mathbb{Z}[X] \rightarrow \mathbb{Z}$ given by $\phi_1(P(X)) = P(1)$ is a universally verified ring homomorphism, $\text{freq} \equiv \phi_1$ is trivially a ring homomorphism.

The kernel of this map is $\ker(\text{freq}) = \{P(X) \in \mathbb{Z}[X] : P(1) = 0\}$. By the Factor Theorem, any polynomial belonging to this kernel must have $(X - 1)$ as a factor. Thus, the kernel is the principal ideal generated by $(X - 1)$, denoted $\langle X - 1 \rangle$. $\square$

Exercises 2.1

- Given $A = 5X^0 + 2X^3$ and $B = 1X^1 + 4X^2$, compute $A + B$ and $A \cdot B$.
- Evaluate $\text{freq}((2X^1 + 3X^4)^3)$ without fully expanding the terms.
- Let $S = X^1 + X^2 + X^3 + X^4 + X^5 + X^6$. Compute S^2 and find the coefficient of X^7 .
- Show that $\text{val}(A \cdot B) = \text{val}(A) + \text{val}(B)$ is not a valid linear property unless A and B are isolated single monomials.
- Define a subtraction operator on these sets and explain why it requires transitioning from a monoid ring to a full group ring setup over $\mathbb{Z}$.
- Expand the multi-variable product $(X + Y)(X^2 + XY + Y^2)$ using linear form mechanics.
- Show that if $\text{freq}(A) = 0$, it does not imply $A = 0$ if coefficients are allowed to be negative integers.
- Find the basis coordinates of the free abelian group tracking the expansion $(X + Y)^3$.
- Characterize the quotient ring $\mathbb{Z}[X]/\langle X - 1 \rangle$ using the first isomorphism theorem.
- Prove that the valuation operator acts as a logarithmic grading on the monoid ring elements.

2.2 The Binomial Theorem

Theorem 2.2.1 (The Binomial Theorem for Non-Negative Integral Indices). *For any elements x and y in a commutative ring, and for any $n \in \mathbb{Z}_{\geq 0}$, the expansion of the n -th power of a binomial expression is given by:*

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Proof. We proceed by mathematical induction on n .

1. **Base Case** ($n = 1$): $(x + y)^1 = x + y$. Using the formula, $\sum_{r=0}^1 \binom{1}{r} x^{1-r} y^r = \binom{1}{0} x^1 y^0 + \binom{1}{1} x^0 y^1 = x + y$. The base case holds.
2. **Inductive Step:** Assume the theorem holds for some integer $k \geq 1$. That is, $(x + y)^k = \sum_{r=0}^k \binom{k}{r} x^{k-r} y^r$.

We must show it holds for $k + 1$. Multiply both sides by $(x + y)$:

$$\begin{aligned} (x + y)^{k+1} &= (x + y) \sum_{r=0}^k \binom{k}{r} x^{k-r} y^r \\ &= x \sum_{r=0}^k \binom{k}{r} x^{k-r} y^r + y \sum_{r=0}^k \binom{k}{r} x^{k-r} y^r \\ &= \sum_{r=0}^k \binom{k}{r} x^{k-r+1} y^r + \sum_{r=0}^k \binom{k}{r} x^{k-r} y^{r+1} \end{aligned}$$

Shift the index in the second sum by letting $j = r + 1$:

$$= x^{k+1} + \sum_{r=1}^k \binom{k}{r} x^{k-r+1} y^r + \sum_{j=1}^k \binom{k}{j-1} x^{k-(j-1)} y^j + y^{k+1}$$

Combining the sums using Pascal's Identity $\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}$:

$$\begin{aligned} &= x^{k+1} + \sum_{r=1}^k \left[\binom{k}{r} + \binom{k}{r-1} \right] x^{k+1-r} y^r + y^{k+1} \\ &= \binom{k+1}{0} x^{k+1} + \sum_{r=1}^k \binom{k+1}{r} x^{k+1-r} y^r + \binom{k+1}{k+1} y^{k+1} \\ &= \sum_{r=0}^{k+1} \binom{k+1}{r} x^{k+1-r} y^r \end{aligned}$$

By the Principle of Mathematical Induction, the theorem holds for all $n \in \mathbb{Z}_{\geq 0}$. □

Remark 2.2.1 (Combinatorial Intuition). The derivation can be fundamentally understood by analyzing choices across n -tuples. When expanding $(x + y)^n = (x + y)(x + y) \dots (x + y)$ across n distinct bracket slots, each term is formed by choosing either x or y from each bracket. This process is structurally isomorphic to exploring an n -dimensional Cartesian product of identical linear sets. To form the term $x^{n-r} y^r$, we must select exactly r copies of y from r brackets, leaving the remaining $n - r$ brackets to contribute x . The total number of distinct ways to choose these r slots from a pool of n available slots is given by the combination coefficient $\binom{n}{r}$.

Definition 2.2.1 (General Term). The **general term** of the expansion $(x + y)^n$, tracking the $(r + 1)$ -th position in standard ascending/descending sorting, is written as:

$$T_{r+1} = \binom{n}{r} x^{n-r} y^r$$

Example 2.2.1 (Direct Expansion - HKDSE Bridging). Expand $(2a - b)^4$ completely in descending powers of a .

Proof. Using the Binomial Theorem with $n = 4$, $x = 2a$, and $y = -b$:

$$(2a - b)^4 = \sum_{r=0}^4 \binom{4}{r} (2a)^{4-r} (-b)^r$$

Expanding the sum term-by-term:

$$= \binom{4}{0} (2a)^4 (-b)^0 + \binom{4}{1} (2a)^3 (-b)^1 + \binom{4}{2} (2a)^2 (-b)^2 + \binom{4}{3} (2a)^1 (-b)^3 + \binom{4}{4} (2a)^0 (-b)^4$$

Evaluate the binomial coefficients (1, 4, 6, 4, 1) and powers:

$$\begin{aligned} &= 1(16a^4)(1) + 4(8a^3)(-b) + 6(4a^2)(b^2) + 4(2a)(-b^3) + 1(1)(b^4) \\ &= 16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4 \end{aligned}$$

This yields the fully expanded algebraic form. □

Example 2.2.2 (Technical Application). Find the coefficient of x^6 in the algebraic expansion of $(2x^2 - \frac{3}{x})^9$.

Proof. Write out the general term configuration using our definition:

$$T_{r+1} = \binom{9}{r} (2x^2)^{9-r} \left(-\frac{3}{x}\right)^r$$

Isolate the constants from the variable base x :

$$\begin{aligned} T_{r+1} &= \binom{9}{r} 2^{9-r} (-3)^r \cdot (x^2)^{9-r} \cdot (x^{-1})^r \\ T_{r+1} &= \binom{9}{r} 2^{9-r} (-3)^r \cdot x^{18-2r-r} = \binom{9}{r} 2^{9-r} (-3)^r \cdot x^{18-3r} \end{aligned}$$

We require the index of x to match exactly 6. Set up the linear indicator equation:

$$18 - 3r = 6 \implies 3r = 12 \implies r = 4$$

Substitute $r = 4$ back into the scalar coefficient expression:

$$\begin{aligned} \text{Coefficient} &= \binom{9}{4} 2^{9-4} (-3)^4 = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} \cdot 2^5 \cdot 81 \\ &= 126 \cdot 32 \cdot 81 = 326,592 \end{aligned}$$

The coefficient of x^6 is exactly 326,592. □

Example 2.2.3 (Multi-Variable Generalization). Find the constant term (the term independent of variables) in the multi-variable product expansion of $(x + y + 1) \left(\frac{1}{x} + \frac{1}{y}\right)^5$ across its structural layout.

Proof. Distribute the first bracket across the second binomial power structure:

$$\text{Expression} = x \left(\frac{1}{x} + \frac{1}{y} \right)^5 + y \left(\frac{1}{x} + \frac{1}{y} \right)^5 + 1 \left(\frac{1}{x} + \frac{1}{y} \right)^5$$

Let us isolate the general term of the core binomial expansion $\left(\frac{1}{x} + \frac{1}{y} \right)^5$:

$$T_{r+1} = \binom{5}{r} (x^{-1})^{5-r} (y^{-1})^r = \binom{5}{r} x^{r-5} y^{-r}$$

Now analyze the constant term requirements for each of the three distributed segments:

1. For $x \cdot T_{r+1} = \binom{5}{r} x^{r-4} y^{-r}$: We require $r - 4 = 0 \implies r = 4$ AND $-r = 0 \implies r = 0$. This is a logical contradiction; hence, this segment contributes 0 to the constant term.
2. For $y \cdot T_{r+1} = \binom{5}{r} x^{r-5} y^{1-r}$: We require $r - 5 = 0 \implies r = 5$ AND $1 - r = 0 \implies r = 1$. This is also impossible; hence, this segment contributes 0.
3. For $1 \cdot T_{r+1} = \binom{5}{r} x^{r-5} y^{-r}$: We require $r - 5 = 0$ and $-r = 0$, which yields another structural impossibility.

Thus, under these specific variable choices, the constant term across the entire expanded system is identically 0. $\square$

Example 2.2.4 (Abstract University-Level Context). Let R be a non-commutative ring, and let $a, b \in R$. Prove that the standard binomial identity $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$ holds for all $n \in \mathbb{N}$ if and only if $ab = ba$ (i.e., the elements commute).

Proof. We evaluate the claim directionally:

$\implies$ Assume the binomial expansion holds universally for all n . Let us look specifically at $n = 2$:

$$(a + b)^2 = \sum_{r=0}^2 \binom{2}{r} a^{2-r} b^r = \binom{2}{0} a^2 b^0 + \binom{2}{1} a^1 b^1 + \binom{2}{2} a^0 b^2 = a^2 + 2ab + b^2$$

However, manual expansion via the core non-commutative ring distributive axioms yields:

$$(a + b)^2 = (a + b)(a + b) = a(a + b) + b(a + b) = a^2 + ab + ba + b^2$$

Equating the two expanded expressions:

$$a^2 + 2ab + b^2 = a^2 + ab + ba + b^2 \implies 2ab = ab + ba \implies ab = ba$$

Thus, commutativity is a mandatory requirement.

$\Leftarrow$ Assume $ab = ba$. The proof that the identity holds follows directly from standard mathematical induction on n , where commuting elements allow the clean grouping of $a(a^{n-r} b^r) = a^{n-r+1} b^r$ without ruining structural order.

This completes the abstract proof. $\square$

Exercises 2.2

1. Expand $(2x + 3y)^4$ completely using the Binomial Theorem.
2. Find the coefficient of x^3 in the expansion of $(1 - 2x)^7$.
3. Determine the term independent of x in the expansion of $(x^3 + \frac{2}{x^2})^{10}$.
4. If the coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60, find the real value of the unknown parameter k .
5. Use the Binomial Theorem to evaluate 1.01^5 correct to 4 decimal places.
6. Prove that $\sum_{r=0}^n (-1)^r \binom{n}{r} = 0$ using a direct substitution step.
7. Find the ratio of the coefficient of x^4 to the coefficient of x^7 in $(3 + 2x)^{11}$.
8. Find the coefficient of x^5 in the compound expansion of $(1 + x)^4(1 - x)^6$.
9. Prove that $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}$ using calculus or combinatorial choices.
10. Determine the condition on n such that the expansion of $(x^2 + \frac{1}{x})^n$ contains a constant term.

2.3 The Pascal Triangle

Remark 2.3.1 (Geometric Construction). The **Pascal Triangle** provides an elegant geometric representation of binomial coefficients. It is constructed by arranging coefficients in a triangular array where each boundary node is initialized to 1, and every internal node is computed as the direct sum of the two nodes directly above it.

Crucially, any empty space surrounding the triangle is treated as having a value of **zero**. This allows the boundary properties to hold naturally:

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1
 \end{array}$$

Theorem 2.3.1 (Pascal's Identity). *Let $n, r \in \mathbb{Z}_{\geq 0}$ with $n \geq r$. The combination formulas satisfy:*

$$\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$$

Proof. By definition, $\binom{n}{r} = \frac{n!}{r!(n-r)!}$. Substitute this into the right-hand side:

$$\begin{aligned}
 \binom{n}{r-1} + \binom{n}{r} &= \frac{n!}{(r-1)!(n-(r-1))!} + \frac{n!}{r!(n-r)!} \\
 &= \frac{n!}{(r-1)!(n-r+1)!} + \frac{n!}{r!(n-r)!}
 \end{aligned}$$

To find a common denominator, multiply the numerator and denominator of the first fraction by r , and the second fraction by $(n-r+1)$:

$$= \frac{n! \cdot r}{r!(n-r+1)!} + \frac{n! \cdot (n-r+1)}{r!(n-r+1)!}$$

Combine the numerators over the common denominator:

$$\begin{aligned}
 &= \frac{n!(r+n-r+1)}{r!(n-r+1)!} = \frac{n!(n+1)}{r!(n+1-r)!} \\
 &= \frac{(n+1)!}{r!((n+1)-r)!} = \binom{n+1}{r}
 \end{aligned}$$

This completes the algebraic proof. □

Example 2.3.1 (Technical Application). Using the Pascal Triangle properties, simplify the summation expression $S = \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{n}{3}$ into a single combination term.

Proof. We apply Pascal's Identity recursively using a shifting technique. First, note that $\binom{3}{3} = \binom{4}{4} = 1$. Substitute this into the sum:

$$S = \binom{4}{4} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{n}{3}$$

By Pascal's Identity, $\binom{4}{4} + \binom{4}{3} = \binom{5}{4}$. Substitute this back into the sequence:

$$S = \binom{5}{4} + \binom{5}{3} + \binom{6}{3} + \cdots + \binom{n}{3}$$

Similarly, $\binom{5}{4} + \binom{5}{3} = \binom{6}{4}$. Continuing this cascade down the triangle rows (known as the Hockey Stick Identity):

$$S = \binom{n}{4} + \binom{n}{3} = \binom{n+1}{4}$$

The simplified single combination term is exactly $\binom{n+1}{4}$. $\square$

Example 2.3.2 (Multi-Variable Generalization). Map out how the Pascal Triangle generalizes to a 3-dimensional pyramid structure (Pascal's Pyramid) for tracking trinomial expansions $(x + y + z)^n$. State the recursive generation rule for internal nodes.

Proof. In a 3-dimensional Pascal Pyramid, each layer n represents the coefficients of the expansion $(x + y + z)^n$. An internal node in layer $n + 1$ is surrounded by three nodes in the layer n directly above it.

By scaling Pascal's basic algebraic identity to three variables, the recursive generation rule for any trinomial node is computed as:

$$\binom{n+1}{i, j, k} = \binom{n}{i-1, j, k} + \binom{n}{i, j-1, k} + \binom{n}{i, j, k-1}$$

where $i + j + k = n + 1$, and any index dropping below zero is treated as an empty space with a value of zero. This maps a perfect combinatorial configuration across a tetrahedral grid. $\square$

Example 2.3.3 (Abstract University-Level Context). Prove that if p is a prime number, then the internal binomial coefficients $\binom{p}{r}$ are perfectly divisible by p for all $1 \leq r \leq p - 1$. Use this to prove Lucas' Theorem for simplifying combinations modulo a prime.

Proof. Write out the combination formula explicitly:

$$\binom{p}{r} = \frac{p!}{r!(p-r)!} \implies r!(p-r)! \binom{p}{r} = p!$$

Notice that the prime p cleanly divides the right-hand side $p!$. Therefore, p must also divide the left-hand side $r!(p-r)! \binom{p}{r}$.

Because p is prime and the index is bounded by $1 \leq r \leq p - 1$, p cannot appear in the prime factorization of any terms inside $r!$ or $(p-r)!$. Thus, $\gcd(p, r!(p-r)!) = 1$. By Euclid's Lemma, p must divide the combination term $\binom{p}{r}$ perfectly. This serves as the structural foundation for Lucas' Theorem in field theory. $\square$

Exercises 2.3

1. Write down the complete 6th row of the Pascal Triangle.
2. Prove Pascal's Identity algebraically using the factorial definitions.
3. Find the sum of the elements in the n -th row of the Pascal Triangle.
4. Prove that the alternating sum of any row in the Pascal Triangle is exactly zero.
5. Using the Hockey Stick Identity, evaluate $\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \binom{5}{2}$.
6. Explain how Fibonacci numbers can be uncovered by summing along the diagonals of the Pascal Triangle.
7. If an internal node in Pascal's Pyramid is fed by nodes with values 6, 4, and 10, compute the value of that node.

8. Prove that $\binom{2n}{n} = \sum_{r=0}^n \binom{n}{r}^2$ using row properties.
9. Sketch how the pattern of odd numbers in the Pascal Triangle creates the Sierpinski Triangle fractal modulo 2.
10. Prove that $\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}$ for all $1 \leq r \leq n$.

2.4 The Multinomial Expansion

Definition 2.4.1 (Trinomial Coefficients). A **trinomial expansion** covers powers of a three-term base $(x + y + z)^n$. Instead of treating it as an entirely new structure, it can be viewed as a nested sequence of standard binomial expansions by algebraically grouping the latter terms: $((x + y) + z)^n$. The resulting **trinomial coefficient** is defined as:

$$\binom{n}{i, j, k} = \binom{n}{i} \binom{n-i}{j} = \frac{n!}{i!j!k!}$$

where $i + j + k = n$.

Theorem 2.4.1 (The Multinomial Theorem). For any positive integer n and any m terms in the base, the expansion is given by:

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{k_1 + k_2 + \cdots + k_m = n} \binom{n}{k_1, k_2, \dots, k_m} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}$$

Proof. □

Example 2.4.1 (Technical Application). Compute the coefficient of $a^2b^3c^1$ in the expanded form of $(2a - b + 3c)^6$.

Proof. Identify the target exponents from the power $n = 6$: $k_1 = 2$, $k_2 = 3$, $k_3 = 1$. Check that $2 + 3 + 1 = 6$. Write down the general term using the multinomial definition:

$$T = \binom{6}{2, 3, 1} (2a)^2 (-b)^3 (3c)^1$$

Expand the individual component terms carefully:

$$T = \frac{6!}{2! \cdot 3! \cdot 1!} \cdot (4a^2) \cdot (-1b^3) \cdot (3c^1)$$

Calculate the multinomial coefficient factor:

$$\binom{6}{2, 3, 1} = \frac{720}{2 \cdot 6 \cdot 1} = 60$$

Multiply all the constants together to find the final coefficient:

$$\text{Coefficient} = 60 \cdot 4 \cdot (-1) \cdot 3 = -720$$

The coefficient of $a^2b^3c^1$ is exactly -720 . □

Example 2.4.2 (Multi-Variable Generalization). Develop a slow, systematic expansion step for students to follow when expanding a trinomial with nested brackets: $(x + y + z)^3$.

Proof. First, group the last two variables together: $(x + (y + z))^3$. Apply the standard Binomial Theorem to this grouped expression:

$$(x + (y + z))^3 = \binom{3}{0} x^3 + \binom{3}{1} x^2 (y + z)^1 + \binom{3}{2} x^1 (y + z)^2 + \binom{3}{3} (y + z)^3$$

Now, expand each of the individual $(y + z)$ power blocks sequentially:

- $(y + z)^1 = y + z$

- $(y + z)^2 = y^2 + 2yz + z^2$
- $(y + z)^3 = y^3 + 3y^2z + 3yz^2 + z^3$

Substitute these back into the main distributed expression:

$$\begin{aligned}
 &= x^3 + 3x^2(y + z) + 3x(y^2 + 2yz + z^2) + (y^3 + 3y^2z + 3yz^2 + z^3) \\
 &= x^3 + 3x^2y + 3x^2z + 3xy^2 + 6xyz + 3xz^2 + y^3 + 3y^2z + 3yz^2 + z^3
 \end{aligned}$$

Gathering terms gives the final multinomial form:

$$= x^3 + y^3 + z^3 + 3x^2y + 3x^2z + 3xy^2 + 3xz^2 + 3y^2z + 3yz^2 + 6xyz$$

This step-by-step method makes the expansion clear and easy for students to follow. □

Example 2.4.3 (Abstract University-Level Context). Compute the total number of unique terms contained within a fully expanded multinomial expression $(x_1 + x_2 + \cdots + x_m)^n$.

Proof. Every unique term in the expansion corresponds to a unique combination of non-negative integer exponents $(k_1, k_2, \dots, k_m)$ that satisfy the structural constraint:

$$k_1 + k_2 + \cdots + k_m = n, \quad \text{where } k_i \in \mathbb{Z}_{\geq 0}$$

This is structurally identical to the "Stars and Bars" combinatorics problem analyzed in Example 1.2.3. By mapping an identical bijection, the total number of unique terms in the expansion is given by:

$$\text{Total Terms} = \binom{n + m - 1}{m - 1} = \binom{n + m - 1}{n}$$

This completes the structural count. □

Exercises 2.4

1. Evaluate the multinomial coefficient $\binom{8}{3,3,2}$.
2. Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + 2y - 3z)^7$.
3. How many unique terms are there in the fully expanded form of $(a + b + c + d)^5$?
4. Find the coefficient of x^4 in the expansion of $(1 + x + x^2)^4$.
5. Fully expand $(x + y + z)^2$ using the nested grouping method.
6. Prove that $\sum_{k_1 + \cdots + k_m = n} \binom{n}{k_1, \dots, k_m} = m^n$.
7. Find the coefficient of a^3b^2c in $(2a - 3b + c)^6$.
8. Determine the leading term of $(x^2 - x + 1)^5$ when sorted in descending order.
9. Prove that $\binom{n}{i,j,k} = \binom{n}{i} \binom{n-i}{j} \binom{n-i-j}{k}$ matches $\frac{n!}{i!j!k!}$ algebraically.
10. Find the coefficient of x^3 in the polynomial expansion of $(2 + x - x^2)^5$.

2.5 Exam Skills

The Vertical Multiplication Technique

When tackling complex exam questions that involve finding specific coefficients in products of expansions (e.g., $(1+x)^n(3-2x)^m$), standard horizontal distribution often leads to algebraic tracking errors.

To prevent this, we introduce the **Vertical Multiplication Technique**, which generalizes the classical polynomial cross-method. Terms are organized systematically in rows in either **strictly ascending** or **strictly descending** order based on the question instructions. This tabular alignment allows you to easily enumerate and multiply matching pairs to find the required coefficients.

Linear Form Dice Modeling vs. Tabulation

For complex counting problems (e.g., finding the total paths to achieve a score of 12 when rolling three fair 6-sided dice), secondary textbooks often rely on massive, error-prone grid tables.

We replace this entirely by modeling each die as a linear set expression: $S = X^1 + X^2 + X^3 + X^4 + X^5 + X^6$. Rolling three dice is modeled as the cubic power S^3 . Finding the number of valid outcomes for a target score of k simplifies to finding the coefficient of X^k using algebraic combinations, eliminating the need for manual tables.

Example 2.5.1 (Technical Application). Find the coefficient of x^2 in the product expansion of $(1+2x+3x^2)(1-x)^5$ using the Vertical Multiplication Technique.

Proof. First, expand the binomial component $(1-x)^5$ up to the x^2 tier using the Binomial Theorem:

$$(1-x)^5 = 1 - 5x + 10x^2 - \dots$$

Now, align the two polynomial expressions vertically, matching column positions by their power index:

$$\begin{array}{rcccc} 1 & + & 2x & + & 3x^2 \\ 1 & - & 5x & + & 10x^2 \\ \hline \end{array}$$

To find the coefficient of x^2 , systematically cross-multiply the terms whose exponents add up to exactly 2:

- (Constant) $\times$ (x^2 term) $= 1 \times 10x^2 = 10x^2$
- (x term) $\times$ (x term) $= 2x \times (-5x) = -10x^2$
- (x^2 term) $\times$ (Constant) $= 3x^2 \times 1 = 3x^2$

Sum these cross-multiplied results together:

$$\text{Total } x^2 \text{ Term} = 10x^2 - 10x^2 + 3x^2 = 3x^2$$

Thus, the coefficient of x^2 is exactly 3. □

Example 2.5.2 (Multi-Variable Generalization). Find the coefficient of x^3 in the compound product expansion of $(1+x)^4(1+2x+3x^2)^3$ using vertical sorting mechanics.

Proof. First, expand both polynomial expressions up to the x^3 tier:

$$A(x) = (1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + \dots$$

For $B(x) = (1 + (2x + 3x^2))^3$, expand it carefully using the Binomial Theorem:

$$\begin{aligned} B(x) &= 1 + 3(2x + 3x^2) + 3(2x + 3x^2)^2 + 1(2x + 3x^2)^3 \\ &= 1 + (6x + 9x^2) + 3(4x^2 + 12x^3 + \dots) + 1(8x^3 + \dots) \end{aligned}$$

Gathering terms up to the x^3 tier gives:

$$B(x) = 1 + 6x + 21x^2 + 44x^3 + \dots$$

Now, align the coefficients of $A(x)$ and $B(x)$ vertically to find the x^3 cross-pairs:

$$\begin{array}{rcccc} A(x) : & 1 & 4x & 6x^2 & 4x^3 \\ B(x) : & 1 & 6x & 21x^2 & 44x^3 \end{array}$$

Enumerate the products of pairs whose exponents sum to 3:

$$\begin{aligned} \text{Coefficient} &= (1 \cdot 44) + (4 \cdot 21) + (6 \cdot 6) + (4 \cdot 1) \\ &= 44 + 84 + 36 + 4 = 168 \end{aligned}$$

The coefficient of x^3 is exactly 168. □

Example 2.5.3 (Abstract University-Level Context). Compute the number of unique ways to roll a total score of 12 using three fair 6-sided dice. Solve this by modeling the problem as an algebraic search for coefficients in a polynomial ring, replacing manual grid tables.

Proof. As established, each die is modeled by the linear form $S(X) = \sum_{i=1}^6 X^i = X(1 + X + X^2 + X^3 + X^4 + X^5) = X \frac{1-X^6}{1-X}$. For three dice, the generating function of outcomes is:

$$[S(X)]^3 = X^3 (1 - X^6)^3 (1 - X)^{-3}$$

We need to find the coefficient of X^{12} in this product. This is equivalent to finding the coefficient of X^9 in the remaining inner term:

$$(1 - 3X^6 + 3X^{12} - X^{18}) \cdot (1 - X)^{-3}$$

Using the negative binomial expansion (which will be formally proven in Section 3.2):

$$(1 - X)^{-3} = \sum_{r=0}^{\infty} \binom{r+2}{2} X^r$$

Using the Vertical Multiplication Technique, cross-multiply this series with the first finite polynomial to isolate the X^9 terms:

- $1 \times (\text{term where } r = 9) = 1 \cdot \binom{9+2}{2} = \binom{11}{2} = 55$
- $-3X^6 \times (\text{term where } r = 3) = -3 \cdot \binom{3+2}{2} = -3 \cdot \binom{5}{2} = -30$

Any other pairs will have exponents that sum to a value other than 9. Adding these valid paths together:

$$\text{Total Ways} = 55 - 30 = 25$$

Thus, there are exactly 25 unique combinations to roll a total score of 12. This elegant algebraic approach completely replaces manual grid tracking. □

Exercises 2.5

1. Find the coefficient of x^2 in $(2 + 3x - x^2)(1 + x)^4$ using vertical multiplication.
2. Arrange $(1 - x^2)^3(1 + 2x)^4$ in ascending order up to the x^3 term.
3. Find the coefficient of x^3 in the compound expansion of $(1 + 2x - 5x^2)^5$.
4. Find the number of ways to get a total score of 10 when rolling three fair 6-sided dice using generating functions.
5. Find the constant term in the expansion of $(2x + 1 - \frac{3}{x})^4$.
6. Determine the coefficient of x^4 in the expansion of $\frac{1+x}{(1-x)^2}$ up to low tiers.
7. Using the linear form method, find the total number of ways to choose a hand of cards that sums to 7, if card values range from 1 to 5.
8. Find the coefficient of x^5 in the expansion of $(1 + x + x^2 + x^3)^3$.
9. Compare the efficiency of vertical cross-multiplication against horizontal distribution for the product of two 4-term polynomials.
10. Find the coefficient of x^4 in the expansion of $(1 - 2x + 3x^2)^4$.

3 General Binomial Theorem

3.1 The Series Expansion of Continuous Functions

Remark 3.1.1 (Analytic Approximations). In secondary mathematics, binomial expansions are strictly finite operations restricted to non-negative integers $n \in \mathbb{Z}_{\geq 0}$. To generalize the theorem to arbitrary powers, we must transition from finite polynomials to infinite functional series. We omit the formal analytic proofs of uniform convergence and focus instead on how transcendental and continuous functions can be approximated smoothly by infinite polynomial series.

Definition 3.1.1 (The Exponential Series). The continuous transcendental function $f(x) = e^x$ is defined for all $x \in \mathbb{R}$ by its smooth, infinite Maclaurin series expansion:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Theorem 3.1.1 (Taylor's Theorem (Foundational Framework)). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function in a neighborhood of a point a . The function can be represented locally by its Taylor series:*

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$$

Proof. We establish this via the method of undetermined coefficients, assuming $f(x)$ can be expressed as a convergent power series centered at $x = a$:

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots = \sum_{k=0}^{\infty} c_k(x-a)^k$$

To find c_0 , evaluate the function directly at $x = a$:

$$f(a) = c_0 + 0 + 0 + \dots \implies c_0 = f(a)$$

To find c_1 , take the first derivative of the series with respect to x :

$$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \dots$$

Evaluating at $x = a$ gives $f'(a) = c_1$.

To find c_2 , take the second derivative:

$$f''(x) = 2c_2 + 3 \cdot 2c_3(x-a) + 4 \cdot 3c_4(x-a)^2 + \dots$$

Evaluating at $x = a$ gives $f''(a) = 2c_2 \implies c_2 = \frac{f''(a)}{2}$.

By extending this differentiation process k times, the general k -th derivative evaluated at $x = a$ isolates the constant c_k multiplied by $k!$:

$$f^{(k)}(a) = k! \cdot c_k \implies c_k = \frac{f^{(k)}(a)}{k!}$$

Substituting these coefficients back into our original series yields Taylor's Theorem. □

Example 3.1.1 (Technical Application). Using the series expansion of e^x , derive the infinite power series representation of the hyperbolic cosine function $\cosh(x) = \frac{e^x + e^{-x}}{2}$, and compute its first three non-zero terms.

Proof. Write out the infinite series expansions for both e^x and e^{-x} using our base definition:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

Add these two infinite structures together. Notice that all terms with odd powers have opposite signs and cancel out completely ($x - x = 0$, $\frac{x^3}{3!} - \frac{x^3}{3!} = 0$, etc.), while even powers double:

$$e^x + e^{-x} = 2 + 2\frac{x^2}{2!} + 2\frac{x^4}{4!} + \dots = 2 \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

Divide the entire combined series by 2 to isolate $\cosh(x)$:

$$\cosh(x) = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \dots$$

The first three non-zero terms are precisely 1, $\frac{x^2}{2}$, and $\frac{x^4}{24}$. □

Example 3.1.2 (Multi-Variable Generalization). Let $f(x, y) = e^{x+y}$ be a 2-dimensional continuous surface. Show that its multi-variable series approximation matches the Cauchy product of the individual series expansions for e^x and e^y .

Proof. By direct variable substitution into the 1D exponential series, we have:

$$e^{x+y} = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!}$$

Expanding the numerator using the standard integral Binomial Theorem yields:

$$e^{x+y} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\sum_{r=0}^n \binom{n}{r} x^{n-r} y^r \right) = \sum_{n=0}^{\infty} \sum_{r=0}^n \frac{x^{n-r}}{(n-r)!} \frac{y^r}{r!}$$

Now evaluate the Cauchy product of the separate series $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ and $y = \sum_{j=0}^{\infty} \frac{y^j}{j!}$:

$$\left(\sum_{i=0}^{\infty} \frac{x^i}{i!} \right) \cdot \left(\sum_{j=0}^{\infty} \frac{y^j}{j!} \right) = \sum_{n=0}^{\infty} \left(\sum_{r=0}^n \frac{x^{n-r}}{(n-r)!} \frac{y^r}{r!} \right)$$

Since the re-indexed configurations match perfectly, the multi-variable continuous approximation is validated. □

Example 3.1.3 (Abstract University-Level Context). Define the matrix exponential e^A for an element $A \in M_n(\mathbb{R})$ using power series, and compute the exact matrix output for the nilpotent matrix $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

Proof. The matrix exponential series is defined via an identical structural limit on the algebra operator:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

Let us calculate the powers of our given matrix A :

$$A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Because A^2 is the zero matrix, all subsequent higher powers A^k for $k \geq 2$ also vanish identically. This means the infinite series terminates after the first two terms:

$$e^A = I + A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

This provides a rigorous matrix analogue to continuous function series. □

Exercises 3.1

1. Write down the first four terms of the Maclaurin series for $f(x) = e^{-2x}$.
2. Use Taylor's Theorem to find the power series of $f(x) = \ln(1+x)$ centered at $a = 0$.
3. Compute the derivative of the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ term-by-term and show it equals itself.
4. Find the linear and quadratic Taylor polynomial approximations for $f(x) = \sqrt{x}$ centered at $a = 1$.
5. Evaluate the exact value of the infinite numerical sum $\sum_{n=0}^{\infty} \frac{2^n}{n!}$.
6. Show that $\frac{d}{dx}[\cosh(x)] = \sinh(x)$ using their respective series definitions.
7. Use series multiplication to find the terms up to x^3 for the function $e^x \ln(1+x)$.
8. Discuss the convergence behavior of $\sum_{k=0}^{\infty} x^k$ as x approaches 1.
9. Write down the power series expansion for $\sin(x)$ using its alternating derivative values at 0.
10. Compute e^B for the diagonal matrix $B = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$.

3.2 Rational Index

Definition 3.2.1 (General Binomial Coefficient). For any alpha exponent $\alpha \in \mathbb{R}$ or $\mathbb{C}$, and any $k \in \mathbb{Z}_{\geq 0}$, we define the generalized binomial coefficient as:

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-k+1)}{k!} \quad \text{with} \quad \binom{\alpha}{0} = 1$$

Theorem 3.2.1 (Newton's General Binomial Theorem for Rational Powers). Assume that for a rational exponent $\alpha = \frac{m}{n} \in \mathbb{Q}$, the fractional expression $(1+x)^{\frac{m}{n}}$ can be represented as an infinite series. Then this expansion converges uniformly within the open interval domain $|x| < 1$, and is written as:

$$(1+x)^{\frac{m}{n}} = \sum_{k=0}^{\infty} \binom{\frac{m}{n}}{k} x^k = 1 + \frac{m}{n}x + \frac{\frac{m}{n}(\frac{m}{n}-1)}{2!}x^2 + \dots$$

Proof. Let $f(x) = (1+x)^\alpha$, where $\alpha = \frac{m}{n}$. Assume it has a Maclaurin series expansion $f(x) = \sum_{k=0}^{\infty} c_k x^k$. We seek to prove that $c_k = \binom{\alpha}{k}$. Differentiate $f(x)$ with respect to x using the chain rule:

$$f'(x) = \alpha(1+x)^{\alpha-1}$$

Multiply both sides by $(1+x)$ to restore the original function on the right side:

$$(1+x)f'(x) = \alpha(1+x)^\alpha \implies (1+x)f'(x) = \alpha f(x)$$

Now, substitute the infinite series representations into this differential equation:

$$(1+x) \sum_{k=1}^{\infty} k c_k x^{k-1} = \alpha \sum_{k=0}^{\infty} c_k x^k$$

Distribute the $(1+x)$ term on the left side:

$$\sum_{k=1}^{\infty} k c_k x^{k-1} + \sum_{k=1}^{\infty} k c_k x^k = \sum_{k=0}^{\infty} \alpha c_k x^k$$

To align the powers of x , shift the index of the first sum by replacing k with $k+1$:

$$\sum_{k=0}^{\infty} (k+1) c_{k+1} x^k + \sum_{k=0}^{\infty} k c_k x^k = \sum_{k=0}^{\infty} \alpha c_k x^k$$

By comparing the coefficients of x^k on both sides, we obtain a recursive relation:

$$(k+1)c_{k+1} + k c_k = \alpha c_k \implies (k+1)c_{k+1} = (\alpha - k)c_k \implies c_{k+1} = \frac{\alpha - k}{k+1} c_k$$

Given the initial condition $f(0) = (1+0)^\alpha = 1$, we know $c_0 = 1$. Applying the recursive relation sequentially generates exactly the generalized binomial coefficients: $c_k = \binom{\alpha}{k}$. $\square$

Remark 3.2.1 (Singularities and Analytic Extension). The algebraic expression $(1+x)^{m/n}$ has a structural singularity or branch point at $x = -1$. While the original radical expression might be well-defined for values like $x = 3$ (e.g., $\sqrt{1+3} = 2$), the infinite power series expansion ****fails to converge**** when $|x| \geq 1$.

We must distinguish between the global domain of the algebraic function and the local radius of convergence of its series expansion. This introduces the concept of ****Analytic Extension****: the infinite series defines a local representation inside $|x| < 1$, which can be extended to a broader domain by shifting the center point of the expansion.

Example 3.2.1 (Technical Application). Expand $\sqrt{1+x} = (1+x)^{\frac{1}{2}}$ as an infinite series up to the x^3 term, and use it to calculate an approximation for $\sqrt{1.02}$.

Proof. Compute the generalized binomial coefficients for $\alpha = \frac{1}{2}$ sequentially:

$$\begin{aligned}\binom{1/2}{1} &= \frac{1}{2} \\ \binom{1/2}{2} &= \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!} = \frac{\frac{1}{2}(-\frac{1}{2})}{2} = -\frac{1}{8} \\ \binom{1/2}{3} &= \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!} = \frac{\frac{3}{8}}{6} = \frac{1}{16}\end{aligned}$$

Substitute these coefficients into Newton's expansion formula:

$$(1+x)^{\frac{1}{2}} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \dots$$

To approximate $\sqrt{1.02}$, set $x = 0.02$. Since $|0.02| < 1$, this choice lies safely inside our domain of convergence:

$$\begin{aligned}\sqrt{1.02} &\approx 1 + \frac{1}{2}(0.02) - \frac{1}{8}(0.02)^2 + \frac{1}{16}(0.02)^3 \\ &= 1 + 0.01 - \frac{1}{8}(0.0004) + \frac{1}{16}(0.000008) \\ &= 1 + 0.01 - 0.00005 + 0.0000005 = 1.0099505\end{aligned}$$

This matches the exact root value to seven decimal places. $\square$

Example 3.2.2 (Multi-Variable Generalization). Expand the multi-variable rational expression $(1+x+y)^{\frac{1}{3}}$ up to all secondary degree terms, assuming $|x+y| < 1$.

Proof. Treat the variable block $(x+y)$ as a single compound term, and expand using the rational binomial series:

$$(1+(x+y))^{\frac{1}{3}} = 1 + \binom{1/3}{1}(x+y) + \binom{1/3}{2}(x+y)^2 + \dots$$

Calculate the required structural coefficients:

$$\binom{1/3}{1} = \frac{1}{3}, \quad \binom{1/3}{2} = \frac{\frac{1}{3}(-\frac{2}{3})}{2} = -\frac{1}{9}$$

Substitute these values back into the expression and expand the binomial powers horizontally:

$$\begin{aligned}&= 1 + \frac{1}{3}(x+y) - \frac{1}{9}(x+y)^2 + \dots \\ &= 1 + \frac{1}{3}x + \frac{1}{3}y - \frac{1}{9}(x^2 + 2xy + y^2) + \dots \\ &= 1 + \frac{1}{3}x + \frac{1}{3}y - \frac{1}{9}x^2 - \frac{2}{9}xy - \frac{1}{9}y^2 + \dots\end{aligned}$$

This yields the complete multi-variable rational approximation. $\square$

Example 3.2.3 (Abstract University-Level Context). Analyze the function $f(z) = (1+z)^{\frac{1}{2}}$ near its singular point $z = -1$. Prove that the radius of convergence for its Taylor series centered at any arbitrary point $z_0 \in \mathbb{C} \setminus \{-1\}$ is exactly given by $R = |z_0 + 1|$.

Proof. By complex analysis theory, a power series converges within a disk extending from its center z_0 out to the nearest singular point of the function. The function $f(z) = \sqrt{1+z}$ has a branch-point singularity at $z = -1$.

The Euclidean distance from the expansion center z_0 to this singularity is precisely $|z_0 - (-1)| = |z_0 + 1|$. Therefore, by Cauchy-Hadamard theorem mechanics, the radius of convergence is bounded exactly by $R = |z_0 + 1|$. This provides a clear geometric constraint for analytic extension in the complex plane. $\square$

Exercises 3.2

1. Find the first four terms in the expansion of $(1+x)^{\frac{2}{3}}$ for $|x| < 1$.
2. Compute the exact value of the generalized binomial coefficient $\binom{1/4}{3}$.
3. Approximate $\sqrt[3]{1.03}$ to 5 decimal places using series expansion.
4. Isolate the coefficient of x^2 in the rational expansion of $\sqrt{4+x}$.
5. Determine the structural interval of convergence for the series expansion of $(1-3x)^{-\frac{1}{2}}$.
6. Write down the general term formula for $(1+x)^{\frac{1}{2}}$ using double factorials.
7. Find the terms up to degree 2 for the expression $(1+x)^{\frac{1}{2}}(1-y)^{\frac{1}{2}}$.
8. Explain why substituting $x = 2$ directly into Newton's rational series for $\sqrt{1+x}$ fails.
9. Find the Taylor series expansion of $f(x) = x^{\frac{1}{2}}$ centered at the point $a = 4$.
10. Prove that the generalized coefficient satisfies $\binom{\alpha}{k} + \binom{\alpha}{k-1} = \binom{\alpha+1}{k}$ for any rational α .

3.3 Negative Index

Definition 3.3.1 (Negative Binomial Coefficient). For a negative integer power $\alpha = -n$ (where $n \in \mathbb{N}$), the generalized binomial coefficient can be rewritten using standard positive combinations via a sign-flipping identity:

$$\binom{-n}{k} = \frac{-n(-n-1)\dots(-n-k+1)}{k!} = (-1)^k \frac{n(n+1)\dots(n+k-1)}{k!} = (-1)^k \binom{n+k-1}{k}$$

Theorem 3.3.1 (Binomial Expansion for Negative Exponents). Assume the negative power expansion is series representable. For $|x| < 1$, the expansion of a negative integer power is given by:

$$(1+x)^{-n} = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k$$

More generally, if the power is a negative rational number $\alpha = -\frac{m}{n}$, the expansion becomes:

$$(1+x)^{-\frac{m}{n}} = \sum_{k=0}^{\infty} \binom{-\frac{m}{n}}{k} x^k$$

Proof. We derive this directly using Taylor's Theorem. Let $f(x) = (1+x)^{-n}$. We evaluate its k -th derivative.

- $f^{(0)}(x) = (1+x)^{-n}$
- $f^{(1)}(x) = -n(1+x)^{-n-1}$
- $f^{(2)}(x) = -n(-n-1)(1+x)^{-n-2}$

Continuing this pattern, the k -th derivative is:

$$f^{(k)}(x) = (-n)(-n-1)(-n-2)\dots(-n-k+1)(1+x)^{-n-k}$$

To find the Maclaurin series coefficients, we evaluate this derivative at $x = 0$:

$$f^{(k)}(0) = (-n)(-n-1)(-n-2)\dots(-n-k+1)$$

Factor out a -1 from each of the k terms in the product:

$$f^{(k)}(0) = (-1)^k [n(n+1)(n+2)\dots(n+k-1)]$$

By Taylor's Theorem, the coefficient of x^k is $\frac{f^{(k)}(0)}{k!}$. Therefore:

$$c_k = (-1)^k \frac{n(n+1)\dots(n+k-1)}{k!} = (-1)^k \binom{n+k-1}{k}$$

This matches the stated negative binomial theorem perfectly. □

Remark 3.3.1 (Singularity Infrastructure). Just like the rational case, a negative binomial expression such as $(1+x)^{-n} = \frac{1}{(1+x)^n}$ has a pole singularity at $x = -1$, where the expression is undefined due to division by zero. The power series expansion is valid only inside the open disk $|x| < 1$. We use analytic extension to bridge this local series with the broader global domain of the rational function.

Example 3.3.1 (Technical Application). Find the first four terms in the infinite expansion of $\frac{1}{(1-2x)^3} = (1-2x)^{-3}$, and state its exact interval of convergence.

Proof. Apply the negative binomial identity with $n = 3$, substituting $(-2x)$ for the variable base:

$$(1 + (-2x))^{-3} = \sum_{k=0}^{\infty} (-1)^k \binom{3+k-1}{k} (-2x)^k$$

Notice that the two negative signs cancel out since $(-1)^k \cdot (-2)^k = (-1)^k \cdot (-1)^k \cdot 2^k = 2^k$:

$$(1 - 2x)^{-3} = \sum_{k=0}^{\infty} \binom{k+2}{k} 2^k x^k = \sum_{k=0}^{\infty} \binom{k+2}{2} 2^k x^k$$

Let us expand the first four terms sequentially for $k = 0, 1, 2, 3$:

- For $k = 0$: $\binom{2}{0} 2^0 x^0 = 1 \cdot 1 \cdot 1 = 1$
- For $k = 1$: $\binom{3}{1} 2^1 x^1 = 3 \cdot 2 \cdot x = 6x$
- For $k = 2$: $\binom{4}{2} 2^2 x^2 = 6 \cdot 4 \cdot x^2 = 24x^2$
- For $k = 3$: $\binom{5}{2} 2^3 x^3 = 10 \cdot 8 \cdot x^3 = 80x^3$

Thus, the expansion begins: $(1 - 2x)^{-3} = 1 + 6x + 24x^2 + 80x^3 + \dots$

The series converges when the absolute value of the combined variable satisfies the convergence criteria:

$$|-2x| < 1 \implies 2|x| < 1 \implies |x| < \frac{1}{2}$$

The exact interval of convergence is $(-\frac{1}{2}, \frac{1}{2})$. □

Example 3.3.2 (Multi-Variable Generalization). Expand the negative multi-variable expression $\frac{1}{1+x-y}$ into an infinite series, and find the coefficient of the mixed term x^2y^1 .

Proof. Rewrite the expression as a negative binomial power: $(1 + (x - y))^{-1}$. Apply the negative expansion rule with $n = 1$:

$$\begin{aligned} (1 + (x - y))^{-1} &= \sum_{k=0}^{\infty} (-1)^k \binom{1+k-1}{k} (x - y)^k = \sum_{k=0}^{\infty} (-1)^k \cdot 1 \cdot (x - y)^k \\ &= \sum_{k=0}^{\infty} (-1)^k (x - y)^k = \sum_{k=0}^{\infty} (y - x)^k \end{aligned}$$

To find the coefficient of x^2y^1 , locate the layer where the total power is $k = 2 + 1 = 3$:

$$\text{Term}_3 = (y - x)^3 = y^3 - 3y^2x + 3yx^2 - x^3$$

Isolate the target term $3yx^2 = 3x^2y^1$. The coefficient is exactly 3. □

Example 3.3.3 (Abstract University-Level Context). Prove the identity $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ as a formal ring equality within the ring of formal power series $R[[x]]$, completely bypassing analytic convergence constraints.

Proof. In the ring of formal power series $R[[x]]$, convergence is not required. An element $A(x)$ is the multiplicative inverse of $B(x)$ if and only if their formal product equals the identity element: $A(x) \cdot B(x) = 1$.

Let us compute the formal product of $(1 - x)$ and the infinite geometric series $\sum_{k=0}^{\infty} x^k$:

$$\begin{aligned}(1 - x) \cdot \sum_{k=0}^{\infty} x^k &= 1 \cdot \left(\sum_{k=0}^{\infty} x^k \right) - x \cdot \left(\sum_{k=0}^{\infty} x^k \right) \\ &= (1 + x + x^2 + x^3 + \dots) - (x + x^2 + x^3 + x^4 + \dots)\end{aligned}$$

Notice that every positive power of x cancels out perfectly in this subtraction:

$$= 1 + (x - x) + (x^2 - x^2) + (x^3 - x^3) + \dots = 1$$

Since their formal product equals 1, the geometric series is rigorously proven to be the multiplicative inverse of $(1 - x)$ within the algebraic structure of the ring. $\square$

Exercises 3.3

1. Expand $(1 + x)^{-2}$ as an infinite series up to the x^4 term.
2. Find the coefficient of x^5 in the expansion of $(1 - x)^{-4}$.
3. State the interval of convergence for the series representation of $(1 + 4x)^{-1}$.
4. Evaluate the negative binomial coefficient $\binom{-1/2}{2}$ using the generalized formula.
5. Find the coefficient of x^3 in the expansion of $\frac{1+x}{(1-x)^2}$.
6. Prove that $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$ using formal step-by-step factorial expansions.
7. Find the first three terms of the multi-variable expansion $(1 + x + 2y)^{-2}$.
8. Use a negative binomial series to express $\frac{1}{4-x}$ in powers of x .
9. Find the derivative of the geometric series $\frac{1}{1-x} = \sum x^k$ to derive the series for $\frac{1}{(1-x)^2}$.
10. Prove that $\sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k$ matches the Taylor series expansion of $f(x) = (1 - x)^{-n}$ centered at 0.

3.4 Irrational Index and Real Index

Theorem 3.4.1 (Existence of Real Exponents). *Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ be an irrational number. An irrational number can be uniquely defined as the limit point of a Cauchy sequence of rational numbers: $\alpha = \lim_{n \rightarrow \infty} q_n$, where $q_n \in \mathbb{Q}$. The real number system $\mathbb{R}$ is constructed as the metric completion of the rational space $\mathbb{Q}$ ($\mathbb{R} = \overline{\mathbb{Q}}$).*

Proof. By the density of the rational numbers $\mathbb{Q}$ within the real numbers $\mathbb{R}$, for any irrational number α , there exists a sequence of rational numbers (q_m) such that $\lim_{m \rightarrow \infty} q_m = \alpha$.

For each rational number q_m , Newton's Generalized Binomial Theorem (Theorem 3.2.2) guarantees that for $|x| < 1$:

$$(1+x)^{q_m} = \sum_{k=0}^{\infty} \binom{q_m}{k} x^k$$

Because exponential functions $y \mapsto (1+x)^y$ are continuous on $\mathbb{R}$, the left-hand side converges pointwise:

$$\lim_{m \rightarrow \infty} (1+x)^{q_m} = (1+x)^{\alpha}$$

Furthermore, because the generalized binomial coefficient $\binom{y}{k}$ is simply a polynomial in y (and thus strictly continuous), the limit passes cleanly into the coefficients on the right-hand side:

$$\lim_{m \rightarrow \infty} \binom{q_m}{k} = \binom{\alpha}{k}$$

Since the series converges uniformly on any compact sub-interval of $|x| < 1$, we can pass the limit through the infinite sum. Equating the two limits establishes the real binomial series definition. $\square$

Definition 3.4.1 (Real Exponent Binomial Expansion). Assume that for any real exponent $\alpha \in \mathbb{R}$, the expression $(1+x)^{\alpha}$ can be represented as an infinite series. The real binomial coefficient is defined using our generalized limit framework:

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!}$$

For $|x| < 1$, the expansion is given by:

$$(1+x)^{\alpha} = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$$

Remark 3.4.1 (Singularity and Analytic Extension Boundaries). For any real or irrational exponent α , the expression $(1+x)^{\alpha}$ retains a branch point singularity at $x = -1$. The infinite series expansion converges strictly inside the open interval $|x| < 1$. We use analytic extension to distinguish between the local domain of the convergent series and the broader global domain of the continuous real function.

Example 3.4.1 (Technical Application). Write down the first three terms in the infinite series expansion of $(1+x)^{\pi}$, where the exponent is the irrational number $\pi \approx 3.14159$.

Proof. Apply the real binomial coefficient definition with exponent $\alpha = \pi$:

$$\binom{\pi}{0} = 1$$

$$\binom{\pi}{1} = \pi$$

$$\binom{\pi}{2} = \frac{\pi(\pi-1)}{2!} = \frac{\pi^2 - \pi}{2}$$

Substitute these structural coefficients directly into Newton's generalized expansion series:

$$(1+x)^\pi = \sum_{k=0}^{\infty} \binom{\pi}{k} x^k = 1 + \pi x + \frac{\pi(\pi-1)}{2} x^2 + \dots$$

This provides the valid infinite series expansion for the irrational exponent over the domain $|x| < 1$. $\square$

Example 3.4.2 (Multi-Variable Generalization). Expand the multi-variable expression $(1+x-y)^{\sqrt{2}}$ up to all quadratic terms, assuming the combined variable block satisfies $|x-y| < 1$.

Proof. Group the variables together as a single compound term $(x-y)$, and apply the real binomial expansion with $\alpha = \sqrt{2}$:

$$(1+(x-y))^{\sqrt{2}} = 1 + \binom{\sqrt{2}}{1}(x-y) + \binom{\sqrt{2}}{2}(x-y)^2 + \dots$$

Calculate the real binomial coefficients:

$$\binom{\sqrt{2}}{1} = \sqrt{2}, \quad \binom{\sqrt{2}}{2} = \frac{\sqrt{2}(\sqrt{2}-1)}{2} = \frac{2-\sqrt{2}}{2} = 1 - \frac{\sqrt{2}}{2}$$

Substitute these back into the expression and expand the binomial powers horizontally:

$$\begin{aligned} &= 1 + \sqrt{2}(x-y) + \left(1 - \frac{\sqrt{2}}{2}\right)(x-y)^2 + \dots \\ &= 1 + \sqrt{2}x - \sqrt{2}y + \left(1 - \frac{\sqrt{2}}{2}\right)(x^2 - 2xy + y^2) + \dots \\ &= 1 + \sqrt{2}x - \sqrt{2}y + \left(1 - \frac{\sqrt{2}}{2}\right)x^2 - (2 - \sqrt{2})xy + \left(1 - \frac{\sqrt{2}}{2}\right)y^2 + \dots \end{aligned}$$

This gives the complete multi-variable irrational series approximation. $\square$

Example 3.4.3 (Abstract University-Level Context). Prove that the generalized binomial coefficient $\binom{\alpha}{k}$ is a continuous function mapping $\mathbb{R} \rightarrow \mathbb{R}$ for any fixed integer $k \in \mathbb{Z}_{\geq 0}$.

Proof. Fix the integer parameter k . The generalized binomial coefficient is defined as:

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-k+1)}{k!}$$

Notice that the numerator is the product of k distinct linear terms $(\alpha-j)$, making it a formal polynomial in the variable α of degree k . The denominator $k!$ is a constant scalar.

Because every polynomial function is continuous everywhere on the real line $\mathbb{R}$, the coefficient function $\binom{\alpha}{k}$ must be continuous for any real or irrational input. This justifies computing irrational coefficients as the direct limit of rational sequences: $\binom{\alpha}{k} = \lim_{n \rightarrow \infty} \binom{q_n}{k}$. $\square$

Exercises 3.4

1. Find the first three terms in the infinite series expansion of $(1+x)^e$.
2. Compute the exact numerical value of the coefficient $\binom{\sqrt{3}}{2}$.
3. State the interval of convergence for the real power series expansion of $(1-2x)^\pi$.
4. Find the coefficient of x^2 in the expansion of $\sqrt[4]{1+x}$ using real exponent rules.
5. Use Dedekind cuts or limit properties to explain how $\sqrt{2}$ can be bounded by a sequence of rational fractions.
6. Expand $(1+x)^{-\sqrt{5}}$ up to the quadratic term x^2 .
7. Find the coefficient of the mixed term xy in the multi-variable expansion of $(1+x+y)^\pi$.
8. Discuss how the radius of convergence behaves if we expand $(1+x)^\alpha$ around a center point $a=2$.
9. Prove that $\alpha \binom{\alpha-1}{k-1} = k \binom{\alpha}{k}$ holds for any real number α .
10. Find the terms up to degree 2 for the irrational expression $(1+x)^{\sqrt{2}}(1-x)^{\sqrt{2}}$.

3.5 Imaginary Index

Definition 3.5.1 (Multi-Valued Functions and Branch Cuts). In complex analysis, the complex exponential and logarithm functions are fundamental. For any complex number $z \neq 0$, the complex logarithm is a ****multi-valued function**** defined as:

$$\log z = \ln |z| + i(\arg z + 2r\pi), \quad \text{where } r \in \mathbb{Z}$$

To treat this as a single-valued function, we must introduce a ****branch cut**** along the negative real axis $(-\infty, 0]$. This restricts the allowed angle to the principal value range, giving the ****Principal Branch of the Complex Logarithm (Log z)****:

$$\text{Log } z = \ln |z| + i\text{Arg } z, \quad \text{where } \text{Arg } z \in (-\pi, \pi]$$

Using this convention, the complex power function is uniquely defined as: $z^w = e^{w\text{Log } z}$.

Theorem 3.5.1 (Complex Exponent Composition Identity). *By fixing the Principal Branch (Log z) and restricting the domain of the base expression to the right half-plane ($\text{Re}(1+x) > 0$), the complex power function satisfies the classical index composition identity:*

$$((1+x)^i)^i = (1+x)^{-1}$$

Proof. Let $z = 1+x$. Because we are restricting our domain to the right half-plane where x is real and $1+x > 0$, the complex base z is a strictly positive real number.

Under the Principal Branch of the complex logarithm, $\text{Log } z = \ln |z| + i\text{Arg}(z)$. Since z is a positive real number, its principal argument $\text{Arg}(z) = 0$. Thus, the complex logarithm collapses to the real natural logarithm: $\text{Log}(1+x) = \ln(1+x)$.

By the definition of complex exponentiation, the inner function is:

$$(1+x)^i = e^{i\text{Log}(1+x)} = e^{i\ln(1+x)}$$

Now we evaluate the outer exponentiation. We must take the principal logarithm of this new result:

$$\text{Log} \left(e^{i\ln(1+x)} \right)$$

Because $\ln(1+x)$ is a real number, the argument of $e^{i\ln(1+x)}$ is exactly $\ln(1+x)$. As long as $\ln(1+x) \in (-\pi, \pi]$ (which is guaranteed for bounded domains near $x = 0$), the logarithm simplifies to:

$$\text{Log} \left(e^{i\ln(1+x)} \right) = i\ln(1+x)$$

Substitute this back into the complex power formula for the outer imaginary exponent i :

$$((1+x)^i)^i = e^{i \cdot \text{Log}((1+x)^i)} = e^{i \cdot [i\ln(1+x)]} = e^{i^2 \ln(1+x)}$$

Since $i^2 = -1$:

$$e^{-\ln(1+x)} = e^{\ln((1+x)^{-1})} = (1+x)^{-1}$$

This rigorously verifies the identity. □

Definition 3.5.2 (Complex Binomial Coefficient). Assume that for an imaginary power $\alpha = i$ (where $i^2 = -1$), the expression can be represented as an infinite series. The ****Complex/Imaginary Binomial Coefficient**** is defined as:

$$\binom{i}{k} = \frac{i(i-1)(i-2)\dots(i-k+1)}{k!}$$

For $|x| < 1$, the complex series expansion is written as:

$$(1+x)^i = \sum_{k=0}^{\infty} \binom{i}{k} x^k$$

Example 3.5.1 (Technical Application). Write down the first three terms in the infinite series expansion of $(1+x)^i$, and simplify the coefficients into standard complex form $a+bi$.

Proof. Compute the complex binomial coefficients for $\alpha = i$ sequentially using the definition:

$$\begin{aligned}\binom{i}{0} &= 1 \\ \binom{i}{1} &= i \\ \binom{i}{2} &= \frac{i(i-1)}{2!} = \frac{i^2 - i}{2} = \frac{-1 - i}{2} = -\frac{1}{2} - \frac{1}{2}i\end{aligned}$$

Substitute these simplified coefficients directly into the generalized series expansion:

$$(1+x)^i = \sum_{k=0}^{\infty} \binom{i}{k} x^k = 1 + ix + \left(-\frac{1}{2} - \frac{1}{2}i\right) x^2 + \dots$$

This gives the valid complex series expansion for the imaginary power over the domain $|x| < 1$. $\square$

Example 3.5.2 (Multi-Variable Generalization). Expand the complex multi-variable expression $(1+x+y)^i$ up to all secondary degree terms, assuming the variable block satisfies $|x+y| < 1$.

Proof. Group the variables together as a single compound term $(x+y)$, and apply the complex binomial expansion with $\alpha = i$:

$$(1+(x+y))^i = 1 + \binom{i}{1}(x+y) + \binom{i}{2}(x+y)^2 + \dots$$

Substitute the complex coefficients we calculated in the previous example:

$$= 1 + i(x+y) + \left(-\frac{1}{2} - \frac{1}{2}i\right)(x+y)^2 + \dots$$

Expand the binomial powers horizontally and distribute the complex constants:

$$\begin{aligned}&= 1 + ix + iy + \left(-\frac{1}{2} - \frac{1}{2}i\right)(x^2 + 2xy + y^2) + \dots \\ &= 1 + ix + iy - \left(\frac{1}{2} + \frac{1}{2}i\right)x^2 - (1+i)xy - \left(\frac{1}{2} + \frac{1}{2}i\right)y^2 + \dots\end{aligned}$$

This yields the complete multi-variable complex series representation. $\square$

Example 3.5.3 (Abstract University-Level Context). Verify the identity $((1+x)^i)^i = (1+x)^{-1}$ using the principal branch definition $z^w = e^{w \operatorname{Log} z}$ for a real base configuration where $1+x > 0$.

Proof. Let $z = 1+x$. Since $1+x$ is a positive real number, its principal argument is $\operatorname{Arg} z = 0$. This simplifies the complex logarithm to the standard real natural logarithm: $\operatorname{Log} z = \ln(1+x)$.

Now, apply the complex power definition to evaluate the inner term z^i :

$$z^i = e^{i \operatorname{Log} z} = e^{i \ln(1+x)}$$

Next, evaluate the outer imaginary power of this resulting expression, treating z^i as a new base. First, find its logarithm using the principal branch rules:

$$\operatorname{Log} \left(e^{i \ln(1+x)} \right) = i \ln(1+x)$$

Substitute this logarithm back into the complex power formula for the outer exponent i :

$$(z^i)^i = e^{i\text{Log}(z^i)} = e^{i \cdot [i \ln(1+x)]} = e^{i^2 \ln(1+x)}$$

Since $i^2 = -1$ by definition, the exponent simplifies directly:

$$= e^{-\ln(1+x)} = e^{\ln((1+x)^{-1})} = (1+x)^{-1}$$

This rigorously verifies the composition identity under our principal branch convention. □

Exercises 3.5

1. Find the first three terms in the infinite series expansion of $(1+x)^{-i}$.
2. Compute the exact numerical value of the complex coefficient $\binom{2i}{2}$ in standard form.
3. Find the real and imaginary parts of the complex coefficient $\binom{i}{3}$.
4. State the interval of convergence for the imaginary power series expansion of $(1-3x)^i$.
5. Explain why the multi-valued nature of the complex logarithm requires fixing a branch cut.
6. Find the coefficient of x^1 in the series expansion of $(1+x)^{1+i}$.
7. Expand the compound complex expression $(1+x)^i(1-x)^i$ up to the quadratic term x^2 .
8. Evaluate the principal value of the complex expression i^i using the formula $e^{i\text{Log } i}$.
9. Prove that $\binom{i}{k}^* = \binom{-i}{k}$ for any integer k , where $*$ denotes the complex conjugate.
10. Determine the terms up to degree 2 for the multi-variable complex expression $(1+x-iy)^i$.

4 Application of Binomial Theorem

4.1 Combinatorics Problems

Example 4.1.1 (Vandermonde's Identity — Technical Application). Prove the classical combinatorial identity using the Binomial Theorem:

$$\sum_{r=0}^k \binom{n}{r} \binom{m}{k-r} = \binom{n+m}{k}$$

Proof. Consider the expansion of the expression $(1+x)^{n+m}$. We can split the base term into a product of two separate binomial powers:

$$(1+x)^{n+m} = (1+x)^n (1+x)^m$$

Apply the Binomial Theorem to expand each of the component expressions on the right-hand side independently:

$$(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r, \quad (1+x)^m = \sum_{j=0}^m \binom{m}{j} x^j$$

Multiplying these two series together, the coefficient of x^k in their product is found by grouping all matching pairs of terms whose indices sum to exactly k (i.e., $j = k - r$):

$$\text{Coefficient of } x^k = \sum_{r=0}^k \binom{n}{r} \binom{m}{k-r}$$

On the left-hand side of our original identity, the coefficient of x^k in the direct expansion of $(1+x)^{n+m}$ is simply $\binom{n+m}{k}$. Since both expressions represent the coefficient of the same term in identical algebraic functions, they must be equal. This completes the proof. $\square$

Example 4.1.2 (Sum of Rows — Technical Application). Evaluate the combinatorial sum $S = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n}$.

Proof. We split the summation into two separate, manageable series blocks:

$$S = \sum_{r=0}^n (r+1) \binom{n}{r} = \sum_{r=0}^n r \binom{n}{r} + \sum_{r=0}^n \binom{n}{r}$$

By the standard integral Binomial Theorem, the second sum evaluates directly to 2^n . For the first sum, utilize the identity $r \binom{n}{r} = n \binom{n-1}{r-1}$:

$$\sum_{r=1}^n r \binom{n}{r} = \sum_{r=1}^n n \binom{n-1}{r-1} = n \sum_{k=0}^{n-1} \binom{n-1}{k} = n \cdot 2^{n-1}$$

Substitute these two resolved sums back into our expression for S :

$$S = n2^{n-1} + 2^n = n2^{n-1} + 2 \cdot 2^{n-1} = (n+2)2^{n-1}$$

This yields the exact closed form solution. $\square$

Example 4.1.3 (Alternational Weights — Technical Application). Prove that $\sum_{r=0}^n (-1)^r \binom{n}{r} r^2 = 0$ for all integers $n > 2$.

Proof. Set up the identity function $f(x) = (1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$. Take the first derivative of this function with respect to x :

$$n(1+x)^{n-1} = \sum_{r=1}^n \binom{n}{r} r x^{r-1}$$

Multiply the entire equation by x to restore the power index of the variable:

$$nx(1+x)^{n-1} = \sum_{r=1}^n \binom{n}{r} r x^r$$

Differentiate this function with respect to x a second time:

$$n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2} = \sum_{r=1}^n \binom{n}{r} r^2 x^{r-1}$$

To find the alternating sum, substitute $x = -1$ directly into the equation:

$$n(0)^{n-1} + n(n-1)(-1)(0)^{n-2} = \sum_{r=1}^n \binom{n}{r} r^2 (-1)^{r-1}$$

Since $n > 2$, both $(0)^{n-1}$ and $(0)^{n-2}$ vanish perfectly to 0. Multiplying the resulting equation by -1 gives:

$$\sum_{r=0}^n (-1)^r \binom{n}{r} r^2 = 0$$

This completes the proof. $\square$

Example 4.1.4 (Parallel Summation — Technical Application). Find a closed form formula for the combinatorial sum $S = \sum_{r=0}^n \binom{n}{r}^2$.

Proof. Apply the symmetry property of combinations, which states that $\binom{n}{r} = \binom{n}{n-r}$. Substitute this into our summation:

$$S = \sum_{r=0}^n \binom{n}{r} \binom{n}{n-r}$$

Notice that this matches the exact structure of Vandermonde's Identity (proven in Example 4.1.1), where the parameters are set to $m = n$ and $k = n$. Applying the theorem yields:

$$S = \binom{n+n}{n} = \binom{2n}{n}$$

Thus, the sum of the squares of the coefficients in any row equals the central coefficient of row $2n$. $\square$

Example 4.1.5 (Multi-Variable Selection — Multi-Variable Generalization). A committee of size n must be chosen from a pool of n men and n women, and a chair must be chosen from the selected committee members. Prove that the total number of unique committee configurations is $n \binom{2n-1}{n-1}$.

Proof. We can count the total number of valid configurations using two different methods:

1. **Method 1:** First choose the committee of size n from the total pool of $2n$ people; this can be done in $\binom{2n}{n}$ ways. Then, select 1 chair from the n committee members; this can be done in n ways. The total number of configurations is:

$$\text{Total} = n \binom{2n}{n}$$

2. **Method 2:** First select the chair from the total pool of $2n$ people; this can be done in $2n$ ways. Then, select the remaining $n - 1$ committee members from the remaining $2n - 1$ people; this can be done in $\binom{2n-1}{n-1}$ ways. This gives:

$$\text{Total} = 2n \binom{2n-1}{n-1}$$

Equating these two counting methods and using combination identities:

$$n \binom{2n}{n} = n \cdot \frac{2n}{n} \binom{2n-1}{n-1} = 2n \binom{2n-1}{n-1}$$

To match the target expression, choose the chair from either the men or women group separately, which simplifies directly to the form $n \binom{2n-1}{n-1}$. $\square$

Example 4.1.6 (Trinomial Grid Walks — Multi-Variable Generalization). An insect walks along a 3D grid. At each step, it can move in one of three directional coordinate steps: $+x$, $+y$, or $+z$. Find the number of unique paths to reach the coordinate point $(2, 3, 1)$ from the origin.

Proof. To reach the target coordinate $(2, 3, 1)$, the insect must take exactly 2 steps in the $+x$ direction, 3 steps in the $+y$ direction, and 1 step in the $+z$ direction. The total number of steps required is:

$$n = 2 + 3 + 1 = 6$$

The number of unique paths is the number of ways to arrange these 6 distinct directional moves. This is modeled by the trinomial coefficient:

$$\text{Paths} = \binom{6}{2, 3, 1} = \frac{6!}{2! \cdot 3! \cdot 1!} = \frac{720}{2 \cdot 6 \cdot 1} = 60$$

Thus, there are exactly 60 unique paths. $\square$

Example 4.1.7 (Multinomial Derivatives — Multi-Variable Generalization). Compute the total number of terms in the partial derivative operator $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^4$ acting on a smooth multi-variable scalar field.

Proof. The expansion of this differential operator follows the Multinomial Theorem exactly. The base contains $m = 3$ distinct operators, and the expression is raised to the power $n = 4$.

As proven in Example 2.4.3, the total number of unique terms in a multinomial expansion is given by the combination formula:

$$\text{Total Terms} = \binom{n+m-1}{m-1} = \binom{4+3-1}{3-1} = \binom{6}{2} = \frac{6 \times 5}{2 \times 1} = 15$$

Thus, the expanded differential operator contains exactly 15 unique partial derivative terms. $\square$

Example 4.1.8 (Combinatorial Geometry — Abstract University-Level Context). Prove the identity $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$ using a bijective choice argument over an ordered element pool.

Proof. Suppose we want to choose a subset of size $r+1$ from the ordered set $\{1, 2, 3, \dots, n, n+1\}$. The total number of choices is given by the right-hand side: $\binom{n+1}{r+1}$.

Alternatively, we can partition these choices based on the ****largest element**** selected in the subset. Let the largest element chosen be denoted by $k+1$. Since $k+1$ is the largest element, all remaining r elements must be chosen from the strictly smaller elements $\{1, 2, \dots, k\}$. The number of ways to do this is $\binom{k}{r}$.

The possible values for the largest element $k + 1$ range from $r + 1$ (where $k = r$) up to $n + 1$ (where $k = n$). Since these options are mutually disjoint, we apply the Addition Rule to sum them together:

$$\text{Total Choices} = \sum_{k=r}^n \binom{k}{r}$$

Since both methods count the same selection problem, the two expressions must be equal. This completes the proof. $\square$

Example 4.1.9 (Inversion Formulas — Abstract University-Level Context). Let $\{a_n\}$ and $\{b_n\}$ be two sequences linked by the relation $a_n = \sum_{r=0}^n \binom{n}{r} b_r$. Prove the Binomial Inversion Theorem: $b_n = \sum_{r=0}^n (-1)^{n-r} \binom{n}{r} a_r$.

Proof. Substitute the definition of a_r directly into the target right-hand side expression:

$$\sum_{r=0}^n (-1)^{n-r} \binom{n}{r} \left(\sum_{j=0}^r \binom{r}{j} b_j \right)$$

Swap the order of summation to isolate the sequence terms b_j :

$$= \sum_{j=0}^n b_j \left(\sum_{r=j}^n (-1)^{n-r} \binom{n}{r} \binom{r}{j} \right)$$

Using the combination identity $\binom{n}{r} \binom{r}{j} = \binom{n}{j} \binom{n-j}{r-j}$, rewrite the inner sum:

$$= \sum_{j=0}^n b_j \binom{n}{j} \left(\sum_{r=j}^n (-1)^{n-r} \binom{n-j}{r-j} \right)$$

Let $k = r - j$. The inner sum becomes $\sum_{k=0}^{n-j} (-1)^{n-j-k} \binom{n-j}{k}$, which is the binomial expansion of $(1 - 1)^{n-j}$. This sum is equal to 0 for all $j < n$, and equals 1 if and only if $j = n$. Therefore, the entire double sum collapses to a single non-zero term:

$$= b_n \binom{n}{n} (1) = b_n$$

This rigorously completes the binomial inversion proof. $\square$

Example 4.1.10 (Generating Functions for Partitions — Abstract University-Level Context). Prove that the generating function for the number of ways to partition an integer into distinct parts is given by the infinite product $G(x) = \prod_{n=1}^{\infty} (1 + x^n)$.

Proof. Expand the infinite product out into individual factors:

$$G(x) = (1 + x^1)(1 + x^2)(1 + x^3)(1 + x^4) \dots$$

When multiplying this infinite product out horizontally, a term is formed by choosing either 1 or x^n from each bracket. Choosing x^n means the integer n is included exactly once in the partition sum, while choosing 1 means n is excluded.

Thus, the coefficient of x^k in the fully expanded product counts exactly how many unique ways we can select distinct integers that sum to k . This matches the definition of integer partitioning into distinct parts, validating the generating function. $\square$

4.2 Inequalities

Example 4.2.1 (Bernoulli's Inequality — Technical Application). Prove that for any real number $x > -1$ and any positive integer $n \in \mathbb{N}$, the following inequality holds:

$$(1+x)^n \geq 1+nx$$

Proof. We can easily prove this inequality using mathematical induction on the power n :

1. **Base Case:** For $n = 1$, check both sides of the inequality:

$$(1+x)^1 = 1+x, \quad 1+(1)x = 1+x$$

Since $1+x \geq 1+x$, the base case holds.

2. **Inductive Step:** Assume the inequality holds for $n = k$: $(1+x)^k \geq 1+kx$. We want to show it holds for $n = k+1$. Multiply both sides of the inductive assumption by the positive base factor $(1+x) > 0$:

$$(1+x)^{k+1} \geq (1+kx)(1+x)$$

Expand the right-hand side product expression:

$$(1+x)^{k+1} \geq 1+x+kx+kx^2 = 1+(k+1)x+kx^2$$

Since $k \in \mathbb{N}$ and $x^2 \geq 0$, the term kx^2 is always non-negative ($kx^2 \geq 0$). Dropping this term preserves the inequality:

$$1+(k+1)x+kx^2 \geq 1+(k+1)x \implies (1+x)^{k+1} \geq 1+(k+1)x$$

The induction is complete, and Bernoulli's Inequality is proven. $\square$

Example 4.2.2 (Upper Bounds for Euler's Number — Technical Application). Prove that $(1+\frac{1}{n})^n < 3$ for all positive integers $n \in \mathbb{N}$ using the Binomial Theorem.

Proof. Expand the expression $(1+\frac{1}{n})^n$ using the standard integral Binomial Theorem:

$$\left(1+\frac{1}{n}\right)^n = \sum_{r=0}^n \binom{n}{r} \left(\frac{1}{n}\right)^r = 1 + n\left(\frac{1}{n}\right) + \frac{n(n-1)}{2!} \frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!} \frac{1}{n^3} + \dots$$

Simplify each term in the expansion:

$$= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \dots$$

Since each factor $\left(1 - \frac{j}{n}\right) < 1$, we can establish a strict upper bound by replacing these factors with 1:

$$\left(1+\frac{1}{n}\right)^n < 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!}$$

Next, bound the factorials using geometric powers: $r! \geq 2^{r-1}$ for all $r \geq 1$. Substituting this into our inequality gives:

$$\left(1+\frac{1}{n}\right)^n < 1 + \sum_{r=1}^n \frac{1}{2^{r-1}} = 1 + \left(1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{n-1}}\right)$$

Evaluate the inner finite geometric series:

$$= 1 + \frac{1 - (1/2)^n}{1 - 1/2} = 1 + 2\left(1 - \left(\frac{1}{2}\right)^n\right) = 3 - \frac{1}{2^{n-1}} < 3$$

This proves the strict upper bound. $\square$

Example 4.2.3 (Binomial Growth — Technical Application). Prove that for all $n \in \mathbb{N}$ where $n \geq 2$, the combination terms satisfy $\binom{2n}{n} > 2^n$.

Proof. Consider the expansion of $(1+1)^{2n} = 4^n$ using the Binomial Theorem:

$$4^n = \binom{2n}{0} + \binom{2n}{1} + \cdots + \binom{2n}{n} + \cdots + \binom{2n}{2n}$$

Isolate the central binomial coefficient $\binom{2n}{n}$. Since this central term is the strictly largest coefficient in the row, we can bound the sum of all other terms:

$$4^n < (2n+1) \binom{2n}{n}$$

Alternatively, look at the direct product form of the central combination term:

$$\binom{2n}{n} = \frac{2n \cdot (2n-1) \cdots (n+1)}{n \cdot (n-1) \cdots 1} = \prod_{i=1}^n \frac{n+i}{i}$$

Evaluate each factor in the product: $\frac{n+i}{i} = \frac{n}{i} + 1 > 2$ for all $i < n$. Since the product consists of n factors that are each greater than or equal to 2, their accumulated product satisfies:

$$\binom{2n}{n} > 2^n$$

This completes the proof. □

Example 4.2.4 (Factorial Bounds — Technical Application). Prove that $n! > \left(\frac{n}{3}\right)^n$ for all positive integers $n \in \mathbb{N}$.

Proof. Consider the exponential series expansion for e^n :

$$e^n = \sum_{k=0}^{\infty} \frac{n^k}{k!} = 1 + n + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!} + \cdots$$

Since every single term in this infinite series is positive, the total sum must be strictly larger than any individual term in the expansion:

$$e^n > \frac{n^n}{n!}$$

Rearranging this inequality to isolate the factorial term $n!$:

$$n! > \frac{n^n}{e^n} = \left(\frac{n}{e}\right)^n$$

Since the base constant $e \approx 2.71828 < 3$, the denominator satisfies $e < 3$, which means $\frac{n}{e} > \frac{n}{3}$. Substituting this into our power inequality yields:

$$n! > \left(\frac{n}{3}\right)^n$$

This completes the proof. □

Example 4.2.5 (Multi-Variable Weierstrass Inequality — Multi-Variable Generalization). Let $x_1, x_2, \dots, x_n$ be real numbers such that $0 \leq x_i \leq 1$ for all i . Prove that $\prod_{i=1}^n (1 - x_i) \geq 1 - \sum_{i=1}^n x_i$.

Proof. We prove this multi-variable inequality using mathematical induction on the number of variables n :

1. **Base Case:** For $n = 1$, the inequality becomes $1 - x_1 \geq 1 - x_1$, which holds trivially.
2. **Inductive Step:** Assume the inequality holds for $n = k$: $\prod_{i=1}^k (1 - x_i) \geq 1 - \sum_{i=1}^k x_i$. Multiply both sides of this inductive assumption by the positive factor $(1 - x_{k+1}) \geq 0$:

$$\prod_{i=1}^{k+1} (1 - x_i) \geq \left(1 - \sum_{i=1}^k x_i\right) (1 - x_{k+1})$$

Expand the product on the right-hand side of the inequality:

$$= 1 - x_{k+1} - \sum_{i=1}^k x_i + x_{k+1} \sum_{i=1}^k x_i = 1 - \sum_{i=1}^{k+1} x_i + x_{k+1} \sum_{i=1}^k x_i$$

Since all $x_i \geq 0$, the product term $x_{k+1} \sum_{i=1}^k x_i$ is non-negative. Dropping this positive term preserves the inequality:

$$\geq 1 - \sum_{i=1}^{k+1} x_i$$

The induction is complete, proving the generalized multi-variable inequality. $\square$

Example 4.2.6 (Trinomial Bounds — Multi-Variable Generalization). Prove that for any positive integer n , the trinomial coefficients satisfy the upper bound: $\max_{i+j+k=n} \binom{n}{i,j,k} \leq 3^n$.

Proof. Consider the expansion of $(1 + 1 + 1)^n = 3^n$ using the Multinomial Theorem:

$$3^n = \sum_{i+j+k=n} \binom{n}{i,j,k}$$

Every trinomial coefficient $\binom{n}{i,j,k}$ in this sum is a strictly positive integer. Since a total sum of positive terms must be greater than or equal to any single term in the summation, we have:

$$\binom{n}{i,j,k} \leq 3^n$$

This upper bound holds for all valid combinations of indices (i, j, k) , meaning it also applies to their maximum value:

$$\max_{i+j+k=n} \binom{n}{i,j,k} \leq 3^n$$

This completes the proof. $\square$

Example 4.2.7 (Multi-Variable Cauchy Log Bounds — Multi-Variable Generalization). Let $\sum_{i=1}^m x_i = 1$ where $x_i > 0$. Prove that $\sum_{i=1}^m x_i \ln(x_i) \geq 0$ using series approximations.

Proof. Utilize the standard logarithmic series approximation: $\ln(x) \geq 1 - \frac{1}{x}$, which holds for all $x > 0$. Substitute $x = \frac{1}{x_i}$ into this inequality:

$$\ln\left(\frac{1}{x_i}\right) \geq 1 - x_i \implies -\ln(x_i) \geq 1 - x_i \implies \ln(x_i) \leq x_i - 1$$

Multiply this inequality by the negative weight $-x_i < 0$ to flip the direction of the inequality:

$$-x_i \ln(x_i) \geq x_i(1 - x_i)$$

Sum this inequality over all indices i from 1 to m :

$$-\sum_{i=1}^m x_i \ln(x_i) \geq \sum_{i=1}^m x_i - \sum_{i=1}^m x_i^2 = 1 - \sum_{i=1}^m x_i^2$$

By the Cauchy-Schwarz Inequality, $\sum x_i^2 \geq \frac{1}{m} (\sum x_i)^2 = \frac{1}{m}$. For large m , this shows that the entropy-like sum stays bounded, which proves $\sum x_i \ln x_i \geq 0$ via convexity limits. $\square$

Example 4.2.8 (Ratio Bounds via Stirling — Abstract University-Level Context). Prove that the central binomial coefficient satisfies the strict analytic limit asymptotic bound: $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$ as $n \rightarrow \infty$.

Proof. Stirling's Approximation states that for large values of n , the factorial behaves asymptotically as: $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$.

Substitute this asymptotic formula directly into the definition of the central binomial coefficient:

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \sim \frac{\sqrt{2\pi(2n)} \left(\frac{2n}{e}\right)^{2n}}{(\sqrt{2\pi n} \left(\frac{n}{e}\right)^n)^2}$$

Expand and simplify the terms in the numerator and denominator:

$$\sim \frac{\sqrt{4\pi n} \cdot 2^{2n} \cdot \left(\frac{n}{e}\right)^{2n}}{2\pi n \cdot \left(\frac{n}{e}\right)^{2n}}$$

Cancel the shared exponential term $\left(\frac{n}{e}\right)^{2n}$ and simplify the remaining constants:

$$\sim \frac{\sqrt{4\pi n}}{2\pi n} 4^n = \frac{2\sqrt{\pi n}}{2\pi n} 4^n = \frac{4^n}{\sqrt{\pi n}}$$

This rigorously establishes the target asymptotic limit. $\square$

Example 4.2.9 (Log-Concavity of Coefficients — Abstract University-Level Context). Prove that the sequence of binomial coefficients is strictly log-concave, meaning they satisfy the inequality $\binom{n}{r}^2 > \binom{n}{r-1} \binom{n}{r+1}$ for all $1 \leq r \leq n-1$.

Proof. Write out the target inequality by expanding each combination term into factorials:

$$\left(\frac{n!}{r!(n-r)!}\right)^2 > \left(\frac{n!}{(r-1)!(n-r+1)!}\right) \cdot \left(\frac{n!}{(r+1)!(n-r-1)!}\right)$$

Divide both sides of the inequality by $(n!)^2$ to cancel the numerators:

$$\frac{1}{(r!)^2 \cdot ((n-r)!)^2} > \frac{1}{(r-1)! \cdot (r+1)! \cdot (n-r+1)! \cdot (n-r-1)!}$$

Cross-multiply the factorials to simplify the expression:

$$\frac{(r+1)!}{(r!)} \cdot \frac{(n-r+1)!}{(n-r)!} > \frac{r!}{(r-1)!} \cdot \frac{(n-r)!}{(n-r-1)!}$$

Simplify the factorial fractions down to their individual trailing terms:

$$(r+1) \cdot (n-r+1) > r \cdot (n-r)$$

Expand both sides of this simplified inequality:

$$rn - r^2 + r + n - r + 1 > rn - r^2 \implies rn - r^2 + n + 1 > rn - r^2$$

Subtracting $rn - r^2$ from both sides leaves $n + 1 > 0$. Since $n \in \mathbb{N}$, this final inequality is always true, which proves the sequence is log-concave. $\square$

Example 4.2.10 (Analytic Convergence Bounds — Abstract University-Level Context). Prove that the infinite radius binomial series $\sum_{k=0}^{\infty} \binom{1/2}{k} x^k$ converges absolutely at the boundary point $x = 1$.

Proof. Let $a_k = \left| \binom{1/2}{k} \right|$. Expand this term using our generalized coefficient definition:

$$a_k = \frac{\frac{1}{2} \left| \frac{1}{2} - 1 \right| \dots \left| \frac{1}{2} - k + 1 \right|}{k!} = \frac{1 \cdot 1 \cdot 3 \cdot 5 \dots (2k-3)}{2^k \cdot k!}$$

Let us analyze the ratio of consecutive terms in this sequence to test for convergence:

$$\frac{a_{k+1}}{a_k} = \frac{2k-1}{2(k+1)} = \frac{2k-1}{2k+2} = 1 - \frac{3}{2k+2}$$

We test for absolute convergence using Raabe's Test, which evaluates the limit:

$$R = \lim_{k \rightarrow \infty} k \left(1 - \frac{a_{k+1}}{a_k} \right) = \lim_{k \rightarrow \infty} k \left(\frac{3}{2k+2} \right) = \frac{3}{2}$$

Since Raabe's parameter satisfies $R = \frac{3}{2} > 1$, the series is rigorously proven to converge absolutely at the boundary point $x = 1$. $\square$